Data Visualization Coursework 2

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1 Decarbonisation and the Electricity Transition in the UK

2 Project Summary

This report examines the decarbonisation of the United Kingdom's electricity system between 2000 and 2025, focusing on the transition from fossil fuel–based generation toward low-carbon sources. Electricity plays a central role in achieving climate targets, and understanding how its composition has evolved is critical for evaluating progress toward a sustainable energy system. Using the Our World in Data – Energy dataset, this study applies exploratory data analysis techniques to investigate changes in the UK's electricity mix, carbon intensity, and broader energy demand patterns.

The analysis explores how the share of electricity generated from fossil fuels has declined over time, how low-carbon sources have expanded and reshaped the generation mix, and whether reductions in energy consumption per capita reflect structural efficiency improvements or broader economic shifts. The findings show a substantial decline in coal's contribution to electricity generation, alongside rapid growth in low-carbon sources, with a key transition point occurring around 2017 when low-carbon electricity surpassed fossil fuels. This shift has contributed to a significant reduction in carbon intensity.

However, the transition remains incomplete. Natural gas continues to play a significant role in electricity generation, highlighting ongoing reliance on fossil fuels. Furthermore, reductions in energy consumption per capita may reflect not only efficiency gains but also structural economic changes, including deindustrialisation. Overall, the findings illustrate both the scale of progress achieved and the challenges that remain in delivering a fully decarbonised and resilient electricity system.

3 Background

Electricity is a fundamental component of modern economies, underpinning industrial activity, digital infrastructure, and everyday life. However, its generation has historically relied on fossil fuels such as coal, oil, and natural gas, making the electricity sector a major contributor to greenhouse gas emissions and environmental degradation (Stern, 2007; IPCC, 2022). As a result, decarbonising electricity systems has become a central objective of global climate policy, and is closely aligned with the United Nations Sustainable Development Goals, particularly SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action).

The United Kingdom has emerged as a leading case in this transition. Through the Climate Change Act and its legally binding commitment to achieve net-zero emissions by 2050, the UK has implemented policies aimed at reducing carbon intensity and transforming its electricity mix (UK Government, 2019; Climate Change Committee, 2023). This has led to a rapid decline in coal-fired generation and a significant expansion of renewable energy, particularly wind power (International Energy Agency, 2022; Ember, 2023).

Despite this progress, the transition is not without challenges. The UK electricity system continues to rely on natural gas to ensure grid stability, highlighting the difficulty of fully eliminating fossil fuels. In addition, reductions in overall energy consumption may not solely reflect efficiency improvements but could also be influenced by broader structural changes, such as deindustrialisation and shifts toward a service-based economy (Barrett et al., 2022; Sovacool et al., 2019). These factors complicate the interpretation of decarbonisation trends and underscore the importance of careful data-driven analysis.

To investigate these dynamics, this report uses the Our World in Data – Energy dataset, which compiles data from sources including the BP Statistical Review of World Energy, the International Energy Agency, and Ember (Our World in Data, 2024). The dataset provides annual country-level observations on electricity generation by source, carbon intensity, and energy consumption, enabling a detailed exploration of the UK's electricity transition between 2000 and 2025.

This study examines how the share of electricity generated from fossil fuels has evolved between 2000 and 2025, with particular attention to how the trajectories of coal and natural gas differ over this period. It further explores how the share of electricity generated from low-carbon sources has changed over time and whether it has overtaken fossil fuels in the overall generation mix. Finally, the analysis investigates how the carbon intensity of electricity has developed, and whether the observed trend reflects a process of structural decarbonisation within the UK power sector, contributing to broader sustainable development objectives.

Table 1: Table 1. Key dataset characteristics.

Attribute	Value
Source	Our World in Data - Energy dataset
Country	United Kingdom
Time period	2000 - 2025
Unit of energy	Terawatt-hours (TWh) unless noted
Observations	26 (one per year)

4 Methodology

The following section presents a series of visualisations designed to explore trends and relationships within the dataset and to answer the research questions outlined above. The analysis proceeds in four stages: package loading, data loading and structural inspection, variable selection and cleaning, and finally descriptive statistics. The visualisations and their interpretations follow in the Results section.

4.1 List of R packages required for this coursework will be loaded

The following packages are required for this analysis:

- tidyverse: A collection of R packages for data manipulation and visualisation, including dplyr, ggplot2, and tidyr. The majority of data wrangling and all visualisations in this work are built on tidyverse functions, making this the most essential dependency.
- ggplot2 (loaded via tidyverse): The primary visualisation engine used throughout this report. Its layered grammar of graphics allows precise control over chart composition, which is particularly important when annotating policy events on time series plots.
- scales: Provides helper functions for formatting axis labels, for example, rendering percentage signs on share variables and comma-separated thousands on consumption figures. Without this package, axis labels would be less readable and less professional in appearance.
- patchwork: Used to combine multiple ggplot2 panels into a single composite figure. This is particularly useful in the preliminary exploration section, where four line plots are arranged in a two-by-two grid to allow direct visual comparison across variables.
- knitr: Required to produce formatted tables within the R Markdown document using the kable() function. This is used to present the dataset characteristics table and the descriptive statistics table in a clean, readable format.
- ggrepel: Used to add non-overlapping text labels to scatter plots. In Visualisation 3, selected year labels are placed adjacent to data points; without ggrepel, these labels would frequently overlap and obscure the underlying data.

```r
library(tidyverse)   # data wrangling + ggplot2
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
```

```
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(scales)      # axis formatting helpers
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(patchwork)   # combining ggplot panels
```

```
## Warning: package 'patchwork' was built under R version 4.5.3
```

```
library(knitr)       # kable tables
library(ggrepel)     # non-overlapping text labels
```

```
## Warning: package 'ggrepel' was built under R version 4.5.3
```

4.2 Data Loading

The dataset is read into R from a local CSV file using read_csv() from the readr package (loaded as part of tidyverse). We load Our World in Data data set in the code chunk below

```
# Load the data set
owid     <- read_csv("owid-energy-data.csv")
```

```
## Rows: 23232 Columns: 130
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (2): country, iso_code
## dbl (128): year, population, gdp, biofuel_cons_change_pct, biofuel_cons_chan...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

To explore the data, there is a need to understand the properties of each of the variables that make up the data set by first taking a glimpse of the data set, which will print the number of rows and columns and brief details of each variable

```r
# To have a glimpse of the dataset
glimpse(owid)
```

```
## Rows: 23,232
## Columns: 130
## $ country                                      <chr> "ASEAN (Ember)", "ASEAN (~
## $ year                                         <dbl> 2000, 2001, 2002, 2003, 2~
## $ iso_code                                     <chr> NA, NA, NA, NA, NA, NA, N~
## $ population                                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gdp                                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_cons_change_pct                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_cons_change_twh                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_cons_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_consumption                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_elec_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_electricity                          <dbl> 5.87, 6.46, 6.62, 7.45, 8~
## $ biofuel_share_elec                           <dbl> 1.550, 1.596, 1.528, 1.62~
## $ biofuel_share_energy                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ carbon_intensity_elec                        <dbl> 572.52, 570.24, 572.68, 5~
## $ coal_cons_change_pct                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_cons_change_twh                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_cons_per_capita                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_consumption                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_elec_per_capita                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_electricity                             <dbl> 76.03, 86.26, 93.43, 102.~
## $ coal_prod_change_pct                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_prod_change_twh                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_prod_per_capita                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_production                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_share_elec                              <dbl> 20.081, 21.307, 21.568, 2~
## $ coal_share_energy                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ electricity_demand                           <dbl> 378.61, 404.85, 433.19, 4~
## $ electricity_demand_per_capita                <dbl> NA, NA, NA, NA, NA, NA, N~
## $ electricity_generation                       <dbl> 378.61, 404.85, 433.19, 4~
## $ electricity_share_energy                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_cons_change_pct                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_cons_change_twh                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_per_capita                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_per_gdp                               <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_cons_change_pct                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_cons_change_twh                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_elec_per_capita                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_electricity                           <dbl> 305.36, 327.66, 356.67, 3~
## $ fossil_energy_per_capita                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_fuel_consumption                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_share_elec                            <dbl> 80.653, 80.934, 82.336, 8~
## $ fossil_share_energy                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_cons_change_pct                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_cons_change_twh                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_consumption                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_elec_per_capita                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_electricity                              <dbl> 164.26, 190.41, 208.92, 2~
## $ gas_energy_per_capita                        <dbl> NA, NA, NA, NA, NA, NA, N~
```

```
## $ gas_prod_change_pct                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_prod_change_twh                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_prod_per_capita                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_production                               <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_share_elec                               <dbl> 43.385, 47.032, 48.228, 4~
## $ gas_share_energy                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ greenhouse_gas_emissions                     <dbl> 216.76, 230.86, 248.08, 2~
## $ hydro_cons_change_pct                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_cons_change_twh                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_consumption                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_elec_per_capita                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_electricity                            <dbl> 50.45, 54.33, 53.29, 53.2~
## $ hydro_energy_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_share_elec                             <dbl> 13.325, 13.420, 12.302, 1~
## $ hydro_share_energy                           <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_cons_change_pct                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_cons_change_twh                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_consumption                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_elec_per_capita                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_electricity                       <dbl> 73.25, 77.19, 76.52, 76.4~
## $ low_carbon_energy_per_capita                 <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_share_elec                        <dbl> 19.347, 19.066, 17.664, 1~
## $ low_carbon_share_energy                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ net_elec_imports                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ net_elec_imports_share_demand                <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_cons_change_pct                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_cons_change_twh                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_consumption                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_elec_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_electricity                          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ nuclear_energy_per_capita                    <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_share_elec                           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ nuclear_share_energy                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_cons_change_pct                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_cons_change_twh                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_consumption                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_elec_per_capita                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_electricity                              <dbl> 65.07, 50.99, 54.32, 53.3~
## $ oil_energy_per_capita                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_prod_change_pct                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_prod_change_twh                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_prod_per_capita                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_production                               <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_share_elec                               <dbl> 17.187, 12.595, 12.540, 1~
## $ oil_share_energy                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewable_consumption                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewable_electricity                  <dbl> 22.80, 22.86, 23.23, 23.1~
## $ other_renewable_exc_biofuel_electricity      <dbl> 16.93, 16.40, 16.61, 15.7~
## $ other_renewables_cons_change_pct             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_cons_change_twh             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_elec_per_capita             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_elec_per_capita_exc_biofuel <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_energy_per_capita           <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_share_elec                  <dbl> 6.022, 5.647, 5.363, 5.06~
```

```
## $ other_renewables_share_elec_exc_biofuel      <dbl> 4.472, 4.051, 3.834, 3.43~
## $ other_renewables_share_energy                <dbl> NA, NA, NA, NA, NA, NA, N~
## $ per_capita_electricity                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ primary_energy_consumption                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_cons_change_pct                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_cons_change_twh                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_consumption                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_elec_per_capita                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_electricity                       <dbl> 73.25, 77.19, 76.52, 76.4~
## $ renewables_energy_per_capita                 <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_share_elec                        <dbl> 19.347, 19.066, 17.664, 1~
## $ renewables_share_energy                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_cons_change_pct                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_cons_change_twh                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_consumption                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_elec_per_capita                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_electricity                            <dbl> 0.00, 0.00, 0.00, 0.00, 0~
## $ solar_energy_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_share_elec                             <dbl> 0.000, 0.000, 0.000, 0.00~
## $ solar_share_energy                           <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_cons_change_pct                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_cons_change_twh                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_consumption                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_elec_per_capita                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_electricity                             <dbl> 0.00, 0.00, 0.00, 0.00, 0~
## $ wind_energy_per_capita                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_share_elec                              <dbl> 0.000, 0.000, 0.000, 0.00~
## $ wind_share_energy                            <dbl> NA, NA, NA, NA, NA, NA, N~
```

Also to get a glimpse of the summary statistics of the respective variables that make up the data set “summary” function will be utilized to print the summary statistics of each of the variables, this will give insights into the distributions of the respective variables.

```r
# To print the summary statistics of the data set
summary(owid)
```

```
##    country               year        iso_code           population       
##  Length:23232       Min.   :1900   Length:23232       Min.   :1.776e+03  
##  Class :character   1st Qu.:1949   Class :character   1st Qu.:1.600e+06  
##  Mode  :character   Median :1985   Mode  :character   Median :6.951e+06  
##                     Mean   :1976                      Mean   :1.058e+08  
##                     3rd Qu.:2005                      3rd Qu.:2.575e+07  
##                     Max.   :2025                      Max.   :8.162e+09  
##                                                       NA's   :4472       
##       gdp            biofuel_cons_change_pct biofuel_cons_change_twh
##  Min.   :1.642e+08   Min.   :-100.000        Min.   :-59.084        
##  1st Qu.:1.426e+10   1st Qu.:  -0.968        1st Qu.:  0.000        
##  Median :4.358e+10   Median :   6.696        Median :  0.000        
##  Mean   :4.258e+11   Mean   :  43.479        Mean   :  1.054        
##  3rd Qu.:1.831e+11   3rd Qu.:  23.321        3rd Qu.:  0.000        
##  Max.   :1.301e+14   Max.   :6373.062        Max.   :144.290        
##  NA's   :11452       NA's   :21162           NA's   :15329          
##  biofuel_cons_per_capita biofuel_consumption biofuel_elec_per_capita
```

```
##  Min.   :   0.0000       Min.   :   0.000    Min.   :   0.000
##  1st Qu.:   0.0000       1st Qu.:   0.000    1st Qu.:   0.000
##  Median :   0.0000       Median :   0.000    Median :   0.874
##  Mean   :  55.1571       Mean   :  15.092    Mean   :  71.065
##  3rd Qu.:   0.5068       3rd Qu.:   0.062    3rd Qu.:  44.114
##  Max.   :2587.7080       Max.   :1366.878    Max.   :2450.791
##  NA's   :16894           NA's   :15175       NA's   :17436
##  biofuel_electricity biofuel_share_elec biofuel_share_energy
##  Min.   :  0.00      Min.   : 0.0000    Min.   :0.0000
##  1st Qu.:  0.00      1st Qu.: 0.0000    1st Qu.:0.0000
##  Median :  0.02      Median : 0.1535    Median :0.0000
##  Mean   : 12.19      Mean   : 1.9626    Mean   :0.2275
##  3rd Qu.:  0.92      3rd Qu.: 1.6082    3rd Qu.:0.0495
##  Max.   :704.81      Max.   :77.5860    Max.   :7.5820
##  NA's   :17125       NA's   :17150      NA's   :16853
##  carbon_intensity_elec coal_cons_change_pct coal_cons_change_twh
##  Min.   :   0.0        Min.   : -100.000    Min.   :-1988.192
##  1st Qu.: 297.0        1st Qu.:   -4.980    1st Qu.:   -1.297
##  Median : 525.3        Median :    1.269    Median :    0.023
##  Mean   : 478.2        Mean   :   22.886    Mean   :   27.702
##  3rd Qu.: 650.3        3rd Qu.:    7.310    3rd Qu.:    6.111
##  Max.   :1306.7        Max.   :55400.000    Max.   : 2811.242
##  NA's   :17126         NA's   :17405        NA's   :16939
##  coal_cons_per_capita coal_consumption    coal_elec_per_capita
##  Min.   :    0.0      Min.   :    0.000   Min.   :   0.0
##  1st Qu.:  304.3      1st Qu.:    4.349   1st Qu.:   0.0
##  Median : 2581.5      Median :   47.364   Median :   0.0
##  Mean   : 5805.0      Mean   : 1682.740   Mean   : 706.2
##  3rd Qu.: 8447.4      3rd Qu.:  446.305   3rd Qu.: 817.6
##  Max.   :96194.4      Max.   :45850.543   Max.   :9421.7
##  NA's   :18087        NA's   :16827       NA's   :16934
##  coal_electricity   coal_prod_change_pct coal_prod_change_twh
##  Min.   :    0.00   Min.   : -100.000    Min.   :-2376.152
##  1st Qu.:    0.00   1st Qu.:   -4.082    1st Qu.:    0.000
##  Median :    0.34   Median :    2.223    Median :    0.000
##  Mean   :  329.05   Mean   :   18.372    Mean   :   17.014
##  3rd Qu.:   36.06   3rd Qu.:    9.101    3rd Qu.:    0.887
##  Max.   :10543.84   Max.   :44965.750    Max.   : 3189.062
##  NA's   :16263      NA's   :13310        NA's   :6670
##  coal_prod_per_capita coal_production     coal_share_elec   coal_share_energy
##  Min.   :     0.00    Min.   :    0.000   Min.   :  0.000   Min.   : 0.000
##  1st Qu.:     0.00    1st Qu.:    0.000   1st Qu.:  0.000   1st Qu.: 2.001
##  Median :    30.58    Median :    1.626   Median :  1.835   Median :12.559
##  Mean   :  3468.44    Mean   :  988.388   Mean   : 17.271   Mean   :18.215
##  3rd Qu.:  2090.02    3rd Qu.:  122.958   3rd Qu.: 32.038   3rd Qu.:29.608
##  Max.   :162212.78    Max.   :50618.328   Max.   :100.000   Max.   :89.157
##  NA's   :8424         NA's   :6410        NA's   :16288     NA's   :16853
##  electricity_demand electricity_demand_per_capita electricity_generation
##  Min.   :    0.00   Min.   :    0.0              Min.   :    0.000
##  1st Qu.:    1.56   1st Qu.:  664.6              1st Qu.:    2.413
##  Median :   12.53   Median : 2610.5              Median :   22.545
##  Mean   :  637.76   Mean   : 4063.9              Mean   :  800.697
##  3rd Qu.:   94.36   3rd Qu.: 5779.9              3rd Qu.:  158.810
##  Max.   :30938.18   Max.   :56048.7              Max.   :30938.180
```

```
##  NA's   :17083      NA's   :17394                NA's   :15548
##  electricity_share_energy energy_cons_change_pct energy_cons_change_twh
##  Min.   : 1.04            Min.   :-100.000       Min.   :-12613.266
##  1st Qu.:12.26            1st Qu.:  -0.525       1st Qu.:    -0.052
##  Median :15.46            Median :   2.483       Median :     0.802
##  Mean   :16.11            Mean   :   4.238       Mean   :    83.496
##  3rd Qu.:18.59            3rd Qu.:   6.301       3rd Qu.:    16.594
##  Max.   :69.00            Max.   :4462.329       Max.   : 10790.422
##  NA's   :19151            NA's   :10487          NA's   :10254
##  energy_per_capita energy_per_gdp   fossil_cons_change_pct
##  Min.   :      0    Min.   : 0.000   Min.   : -49.5910
##  1st Qu.:  2833    1st Qu.: 0.769   1st Qu.:  -0.6685
##  Median : 12667    Median : 1.250   Median :   2.6230
##  Mean   : 25909    Mean   : 1.686   Mean   :   3.2784
##  3rd Qu.: 36636    3rd Qu.: 2.066   3rd Qu.:   5.9710
##  Max.   :651837    Max.   :25.253   Max.   :1553.1050
##  NA's   :12146     NA's   :15350    NA's   :16965
##  fossil_cons_change_twh fossil_elec_per_capita fossil_electricity
##  Min.   :-9069.922      Min.   :    0.0        Min.   :    0.00
##  1st Qu.:   -1.397      1st Qu.:  254.3        1st Qu.:    0.64
##  Median :    7.005      Median : 1418.8        Median :   10.09
##  Mean   :   96.807      Mean   : 2464.3        Mean   :  550.98
##  3rd Qu.:   41.473      3rd Qu.: 3533.7        3rd Qu.:  127.57
##  Max.   : 9430.391      Max.   :23215.5        Max.   :18313.96
##  NA's   :16939          NA's   :16922          NA's   :16251
##  fossil_energy_per_capita fossil_fuel_consumption fossil_share_elec
##  Min.   :     0           Min.   :     0.0       Min.   :  0.00
##  1st Qu.: 10360           1st Qu.:   142.4       1st Qu.: 42.83
##  Median : 24975           Median :   406.5       Median : 70.95
##  Mean   : 31749           Mean   :  5271.0       Mean   : 65.06
##  3rd Qu.: 40052           3rd Qu.:  1870.4       3rd Qu.: 94.29
##  Max.   :318560           Max.   :142420.9       Max.   :100.00
##  NA's   :18087            NA's   :16827          NA's   :16276
##  fossil_share_energy gas_cons_change_pct gas_cons_change_twh gas_consumption
##  Min.   : 13.87      Min.   :  -95.245   Min.   :-3749.125   Min.   :    0.00
##  1st Qu.: 80.82      1st Qu.:   -0.557   1st Qu.:    0.000   1st Qu.:   11.88
##  Median : 90.36      Median :    4.083   Median :    1.872   Median :   82.81
##  Mean   : 86.07      Mean   :   18.526   Mean   :   33.585   Mean   : 1284.21
##  3rd Qu.: 96.66      3rd Qu.:    9.989   3rd Qu.:   16.274   3rd Qu.:  440.57
##  Max.   :100.00      Max.   :15299.998   Max.   : 3625.422   Max.   :41278.27
##  NA's   :16853       NA's   :17589       NA's   :16939       NA's   :16827
##  gas_elec_per_capita gas_electricity   gas_energy_per_capita
##  Min.   :    0.00    Min.   :   0.00   Min.   :     0.0
##  1st Qu.:    0.00    1st Qu.:   0.00   1st Qu.:   613.1
##  Median :   89.81    Median :   2.51   Median :  4266.0
##  Mean   :  914.96    Mean   : 183.18   Mean   :  9977.0
##  3rd Qu.:  853.34    3rd Qu.:  46.40   3rd Qu.: 10490.1
##  Max.   :23215.47    Max.   :6900.36   Max.   :288358.5
##  NA's   :17050       NA's   :16379     NA's   :18087
##  gas_prod_change_pct  gas_prod_change_twh gas_prod_per_capita
##  Min.   :-1.000e+02   Min.   :-4092.556   Min.   :      0.0
##  1st Qu.:-1.000e+00   1st Qu.:    0.000   1st Qu.:      0.0
##  Median : 4.000e+00   Median :    0.000   Median :      0.0
##  Mean   : 1.151e+14   Mean   :   17.562   Mean   :   6655.4
```

```
##  3rd Qu.: 1.100e+01   3rd Qu.:    1.348   3rd Qu.:   770.6
##  Max.   : 9.323e+17   Max.   : 4146.363   Max.   :852312.1
##  NA's   :15129        NA's   :6242        NA's   :8204
##  gas_production     gas_share_elec   gas_share_energy greenhouse_gas_emissions
##  Min.   :    0.00   Min.   :  0.00   Min.   : 0.000   Min.   :    0.000
##  1st Qu.:    0.00   1st Qu.:  0.00   1st Qu.: 4.941   1st Qu.:    0.300
##  Median :    0.00   Median :  9.85   Median :16.468   Median :    3.365
##  Mean   :  625.12   Mean   : 19.66   Mean   :20.096   Mean   :  339.127
##  3rd Qu.:   62.22   3rd Qu.: 27.52   3rd Qu.:28.062   3rd Qu.:   41.045
##  Max.   :41244.52   Max.   :100.00   Max.   :91.942   Max.   :14597.910
##  NA's   :5981       NA's   :16402    NA's   :16853    NA's   :17470
##  hydro_cons_change_pct hydro_cons_change_twh hydro_consumption
##  Min.   : -81.250      Min.   :-441.637      Min.   :    0.000
##  1st Qu.:  -3.890      1st Qu.:  -0.519      1st Qu.:    3.517
##  Median :   2.818      Median :   0.128      Median :   28.391
##  Mean   :   5.827      Mean   :   7.817      Mean   :  379.975
##  3rd Qu.:  10.734      3rd Qu.:   4.757      3rd Qu.:  157.449
##  Max.   :1750.000      Max.   : 485.917      Max.   :10860.753
##  NA's   :17546         NA's   :16939         NA's   :16827
##  hydro_elec_per_capita hydro_electricity hydro_energy_per_capita
##  Min.   :    0.000     Min.   :   0.00   Min.   :     0.0
##  1st Qu.:    6.971     1st Qu.:   0.10   1st Qu.:   121.1
##  Median :  140.595     Median :   3.72   Median :   627.6
##  Mean   :  983.781     Mean   : 121.46   Mean   :  3449.3
##  3rd Qu.:  710.102     3rd Qu.:  30.86   3rd Qu.:  2269.7
##  Max.   :40911.043     Max.   :4419.95   Max.   :105414.0
##  NA's   :14946         NA's   :13855     NA's   :18087
##  hydro_share_elec   hydro_share_energy low_carbon_cons_change_pct
##  Min.   :  0.0000   Min.   : 0.000     Min.   : -88.2080
##  1st Qu.:  0.6168   1st Qu.: 1.105     1st Qu.:  -0.6042
##  Median : 13.6780   Median : 5.361     Median :   4.3640
##  Mean   : 25.1029   Mean   : 9.338     Mean   :  11.3946
##  3rd Qu.: 41.9147   3rd Qu.:11.989     3rd Qu.:  11.1808
##  Max.   :100.0000   Max.   :71.417     Max.   :7518.2040
##  NA's   :15680      NA's   :16853      NA's   :17330
##  low_carbon_cons_change_twh low_carbon_consumption low_carbon_elec_per_capita
##  Min.   :-669.799           Min.   :    0.00       Min.   :    0.0
##  1st Qu.:  -0.028           1st Qu.:    7.05       1st Qu.:   30.3
##  Median :   1.014           Median :   51.77       Median :  261.9
##  Mean   :  29.272           Mean   :  793.53       Mean   : 1507.8
##  3rd Qu.:  10.937           3rd Qu.:  263.19       3rd Qu.: 1415.4
##  Max.   :2359.283           Max.   :32850.86       Max.   :56048.7
##  NA's   :16939              NA's   :16827          NA's   :14852
##  low_carbon_electricity low_carbon_energy_per_capita low_carbon_share_elec
##  Min.   :    0.00       Min.   :     0.0             Min.   :  0.000
##  1st Qu.:    0.25       1st Qu.:   259.6             1st Qu.:  5.663
##  Median :    6.27       Median :  1480.6             Median : 29.614
##  Mean   :  258.75       Mean   :  5723.0             Mean   : 35.530
##  3rd Qu.:   49.49       3rd Qu.:  5511.8             3rd Qu.: 59.340
##  Max.   :12624.22       Max.   :153154.0             Max.   :100.000
##  NA's   :13761          NA's   :18087                NA's   :15586
##  low_carbon_share_energy net_elec_imports   net_elec_imports_share_demand
##  Min.   : 0.000          Min.   :-93.3200   Min.   :-772.594
##  1st Qu.: 3.336          1st Qu.:  0.0000   1st Qu.:   0.000
```

```
##  Median : 9.635         Median :  0.0000   Median :   0.000
##  Mean   :13.929         Mean   :  0.1129   Mean   :   1.087
##  3rd Qu.:19.181         3rd Qu.:  0.5250   3rd Qu.:   2.236
##  Max.   :86.126         Max.   : 66.6700   Max.   :  98.690
##  NA's   :16853          NA's   :17393      NA's   :17416
##  nuclear_cons_change_pct nuclear_cons_change_twh nuclear_consumption
##  Min.   : -100.000       Min.   :-534.551        Min.   :   0.00
##  1st Qu.:   -1.425       1st Qu.:   0.000        1st Qu.:   0.00
##  Median :    3.320       Median :   0.000        Median :   0.00
##  Mean   :   32.429       Mean   :   5.482        Mean   : 228.03
##  3rd Qu.:   14.155       3rd Qu.:   0.000        3rd Qu.:  25.77
##  Max.   :11200.001       Max.   : 875.458        Max.   :7494.95
##  NA's   :20653           NA's   :15515           NA's   :15373
##  nuclear_elec_per_capita nuclear_electricity nuclear_energy_per_capita
##  Min.   :   0.0          Min.   :   0.000    Min.   :    0.0
##  1st Qu.:   0.0          1st Qu.:   0.000    1st Qu.:    0.0
##  Median :   0.0          Median :   0.000    Median :    0.0
##  Mean   : 313.6          Mean   :  90.840    Mean   : 1123.5
##  3rd Qu.:   0.0          3rd Qu.:   3.977    3rd Qu.:  150.7
##  Max.   :8908.0          Max.   :2778.760    Max.   :24744.8
##  NA's   :14260           NA's   :13169       NA's   :16647
##  nuclear_share_elec nuclear_share_energy oil_cons_change_pct
##  Min.   : 0.000     Min.   : 0.000       Min.   : -73.921
##  1st Qu.: 0.000     1st Qu.: 0.000       1st Qu.:  -1.194
##  Median : 0.000     Median : 0.000       Median :   2.386
##  Mean   : 5.815     Mean   : 2.811       Mean   :   3.080
##  3rd Qu.: 2.248     3rd Qu.: 2.137       3rd Qu.:   6.423
##  Max.   :88.011     Max.   :41.661       Max.   :1553.105
##  NA's   :15677      NA's   :16853        NA's   :16965
##  oil_cons_change_twh oil_consumption    oil_elec_per_capita oil_electricity
##  Min.   :-4896.020   Min.   :    0.00   Min.   :    0.00    Min.   :   0.00
##  1st Qu.:   -1.666   1st Qu.:   80.39   1st Qu.:   26.77    1st Qu.:   0.15
##  Median :    3.615   Median :  204.00   Median :  157.29    Median :   1.28
##  Mean   :   35.520   Mean   : 2304.06   Mean   :  891.33    Mean   :  45.20
##  3rd Qu.:   20.542   3rd Qu.: 1013.22   3rd Qu.:  573.54    3rd Qu.:  12.78
##  Max.   : 2993.727   Max.   :55292.08   Max.   :14645.94    Max.   :1366.09
##  NA's   :16939       NA's   :16827      NA's   :16983       NA's   :16312
##  oil_energy_per_capita oil_prod_change_pct oil_prod_change_twh
##  Min.   :     0        Min.   : -100.000   Min.   :-6455.246
##  1st Qu.:  5098        1st Qu.:   -3.092   1st Qu.:    0.000
##  Median : 11817        Median :    1.756   Median :    0.000
##  Mean   : 15967        Mean   :   21.801   Mean   :   23.929
##  3rd Qu.: 21161        3rd Qu.:   10.198   3rd Qu.:    1.756
##  Max.   :156815        Max.   :26472.602   Max.   : 5856.434
##  NA's   :18087         NA's   :13050       NA's   :5503
##  oil_prod_per_capita oil_production      oil_share_elec    oil_share_energy
##  Min.   :      0.0   Min.   :    0.000   Min.   :  0.000   Min.   :  8.058
##  1st Qu.:      0.0   1st Qu.:    0.000   1st Qu.:  2.033   1st Qu.: 33.958
##  Median :      5.6   Median :    1.896   Median :  8.633   Median : 43.903
##  Mean   :  25975.0   Mean   : 1230.348   Mean   : 29.034   Mean   : 47.759
##  3rd Qu.:   2450.1   3rd Qu.:  185.722   3rd Qu.: 50.010   3rd Qu.: 60.315
##  Max.   :3443292.2   Max.   :52831.051   Max.   :100.000   Max.   :100.000
##  NA's   :7658        NA's   :5240        NA's   :16337     NA's   :16853
##  other_renewable_consumption other_renewable_electricity
```

```
##  Min.   :   0.000           Min.   :  0.00
##  1st Qu.:   0.000           1st Qu.:  0.00
##  Median :   0.555           Median :  0.03
##  Mean   :  42.553           Mean   : 13.21
##  3rd Qu.:  11.225           3rd Qu.:  1.65
##  Max.   :2476.357           Max.   :791.79
##  NA's   :16827              NA's   :13792
##  other_renewable_exc_biofuel_electricity other_renewables_cons_change_pct
##  Min.   : 0.00                           Min.   : -100.0000
##  1st Qu.: 0.00                           1st Qu.:   -0.2708
##  Median : 0.00                           Median :    5.0550
##  Mean   : 2.08                           Mean   :   32.7613
##  3rd Qu.: 0.00                           3rd Qu.:   13.7608
##  Max.   :87.91                           Max.   :21131.3340
##  NA's   :17700                           NA's   :19244
##  other_renewables_cons_change_twh other_renewables_elec_per_capita
##  Min.   :-48.202                  Min.   :    0.000
##  1st Qu.:  0.000                  1st Qu.:    0.000
##  Median :  0.000                  Median :    0.226
##  Mean   :  2.369                  Mean   :  103.615
##  3rd Qu.:  0.441                  3rd Qu.:   39.173
##  Max.   :165.264                  Max.   :17032.252
##  NA's   :16939                    NA's   :14883
##  other_renewables_elec_per_capita_exc_biofuel
##  Min.   :    0.00
##  1st Qu.:    0.00
##  Median :    0.00
##  Mean   :   70.08
##  3rd Qu.:    0.00
##  Max.   :17032.25
##  NA's   :18011
##  other_renewables_energy_per_capita other_renewables_share_elec
##  Min.   :    0.00                   Min.   : 0.000
##  1st Qu.:    0.00                   1st Qu.: 0.000
##  Median :   19.93                   Median : 0.258
##  Mean   :  470.59                   Mean   : 2.412
##  3rd Qu.:  224.51                   3rd Qu.: 1.850
##  Max.   :53224.31                   Max.   :77.586
##  NA's   :18087                      NA's   :15617
##  other_renewables_share_elec_exc_biofuel other_renewables_share_energy
##  Min.   : 0.0000                         Min.   : 0.0000
##  1st Qu.: 0.0000                         1st Qu.: 0.0000
##  Median : 0.0000                         Median : 0.1180
##  Mean   : 0.7591                         Mean   : 0.8148
##  3rd Qu.: 0.0000                         3rd Qu.: 0.7060
##  Max.   :48.4500                         Max.   :31.2440
##  NA's   :17725                           NA's   :16853
##  per_capita_electricity primary_energy_consumption renewables_cons_change_pct
##  Min.   :    0.0        Min.   :0.000e+00          Min.   : -92.641
##  1st Qu.:  659.9        1st Qu.:8.002e+00          1st Qu.:  -1.338
##  Median : 2587.7        Median :9.119e+01          Median :   4.529
##  Mean   : 4054.6        Mean   :4.255e+03          Mean   :  11.091
##  3rd Qu.: 5773.8        3rd Qu.:7.722e+02          3rd Qu.:  12.001
##  Max.   :56048.7        Max.   :1.767e+05          Max.   :7518.204
```

```
##  NA's   :16379          NA's   :9977              NA's   :17337
##  renewables_cons_change_twh renewables_consumption renewables_elec_per_capita
##  Min.   :-430.247           Min.   :    0.000      Min.   :    0.00
##  1st Qu.:  -0.061           1st Qu.:    5.299      1st Qu.:   28.21
##  Median :   0.701           Median :   37.118      Median :  216.60
##  Mean   :  22.549           Mean   :  513.739      Mean   : 1172.04
##  3rd Qu.:   7.878           3rd Qu.:  190.562      3rd Qu.:  922.20
##  Max.   :2164.578           Max.   :25979.023      Max.   :56048.73
##  NA's   :16939              NA's   :16827          NA's   :14852
##  renewables_electricity renewables_energy_per_capita renewables_share_elec
##  Min.   :   0.000       Min.   :     0.0             Min.   :  0.000
##  1st Qu.:   0.223       1st Qu.:   208.4             1st Qu.:  3.514
##  Median :   4.610       Median :   914.9             Median : 18.500
##  Mean   : 162.252       Mean   :  4285.0             Mean   : 29.787
##  3rd Qu.:  36.724       3rd Qu.:  3235.7             3rd Qu.: 50.000
##  Max.   :9845.460       Max.   :153154.0             Max.   :100.000
##  NA's   :13761          NA's   :18087                NA's   :15586
##  renewables_share_energy solar_cons_change_pct solar_cons_change_twh
##  Min.   : 0.000          Min.   : -100.00      Min.   :  -4.592
##  1st Qu.: 2.127          1st Qu.:   10.56      1st Qu.:   0.000
##  Median : 6.551          Median :   28.41      Median :   0.000
##  Mean   :11.118          Mean   :  124.40      Mean   :   5.018
##  3rd Qu.:15.008          3rd Qu.:   64.15      3rd Qu.:   0.014
##  Max.   :86.126          Max.   :21800.00      Max.   :1123.988
##  NA's   :16853           NA's   :20717         NA's   :16939
##  solar_consumption  solar_elec_per_capita solar_electricity
##  Min.   :   0.000   Min.   :   0.000      Min.   :   0.000
##  1st Qu.:   0.000   1st Qu.:   0.000      1st Qu.:   0.000
##  Median :   0.000   Median :   0.000      Median :   0.000
##  Mean   :  23.626   Mean   :  28.961      Mean   :   9.275
##  3rd Qu.:   0.094   3rd Qu.:   0.607      3rd Qu.:   0.024
##  Max.   :5150.569   Max.   :1879.595      Max.   :2128.370
##  NA's   :16827      NA's   :14838         NA's   :13747
##  solar_energy_per_capita solar_share_elec  solar_share_energy
##  Min.   :   0.000        Min.   : 0.0000   Min.   :0.0000
##  1st Qu.:   0.000        1st Qu.: 0.0000   1st Qu.:0.0000
##  Median :   0.000        Median : 0.0000   Median :0.0000
##  Mean   :  82.237        Mean   : 0.9701   Mean   :0.2198
##  3rd Qu.:   2.442        3rd Qu.: 0.2148   3rd Qu.:0.0060
##  Max.   :4584.596        Max.   :50.0000   Max.   :9.8810
##  NA's   :18087           NA's   :15592     NA's   :16853
##  wind_cons_change_pct wind_cons_change_twh wind_consumption
##  Min.   :  -100.000   Min.   :-46.535      Min.   :   0.000
##  1st Qu.:      2.329   1st Qu.:  0.000      1st Qu.:   0.000
##  Median :     18.349   Median :  0.000      Median :   0.000
##  Mean   :    290.408   Mean   :  6.021      Mean   :  48.691
##  3rd Qu.:     45.211   3rd Qu.:  0.096      3rd Qu.:   0.916
##  Max.   :242384.859   Max.   :642.617      Max.   :6124.465
##  NA's   :20434        NA's   :16939        NA's   :16827
##  wind_elec_per_capita wind_electricity  wind_energy_per_capita wind_share_elec
##  Min.   :   0.000     Min.   :   0.00   Min.   :   0.00        Min.   : 0.000
##  1st Qu.:   0.000     1st Qu.:   0.00   1st Qu.:   0.00        1st Qu.: 0.000
##  Median :   0.000     Median :   0.00   Median :   0.00        Median : 0.000
##  Mean   :  68.159     Mean   :  19.81   Mean   : 214.96        Mean   : 1.655
```

```
##  3rd Qu.:   1.855     3rd Qu.:   0.12   3rd Qu.:  25.91        3rd Qu.: 0.565
##  Max.   :3947.840     Max.   :2505.35   Max.   :9315.29        Max.   :58.357
##  NA's   :14995        NA's   :13904     NA's   :18087          NA's   :15749
##  wind_share_energy
##  Min.   : 0.0000
##  1st Qu.: 0.0000
##  Median : 0.0000
##  Mean   : 0.5181
##  3rd Qu.: 0.0745
##  Max.   :26.0850
##  NA's   :16853
```

The above code chunk has revealed the type of variables that make up the Our World in Data data set, the missing values and the summary statistics of each numerical variable. From the code chunk, it was observed that the number of columns and variables in this data is 130, while the number of rows is 23,232.

```
# Sum of missing values
sum(is.na(owid))
```

```
## [1] 1988709
```

It was also observed that the data set has 1988709 missing values which will be dealt with in various sections of this course work.

From a careful analysis of each variables' summary statistics and characteristics and their attributes, it is evident that answering the questions below and as captured in the introduction about the Decarbonisation and the Electricity Transition in the UK is reasonably possible.

1. How has the share of electricity generated from fossil fuels changed between 2000 and 2025, and how do the individual trajectories of coal and natural gas compare during this transition?
2. How has the share of electricity generated from low-carbon sources changed over time, has it surpassed fossil fuels, and which technologies have driven this transition?
3. How has the carbon intensity of electricity evolved over time, and does the trend indicate a structural decarbonisation of the UK power sector?

The full Our World in Data Energy dataset covers all countries, we will therefore need to start by filtering the variables to only the United Kingdom since that is what our work will be focusing on.

```
# Filter to United Kingdom
uk_data  <- owid %>% filter(country == "United Kingdom")
# To get a glimpse of the uk_data
glimpse(uk_data)
```

```
## Rows: 126
## Columns: 130
## $ country                                        <chr> "United Kingdom", "United~
## $ year                                           <dbl> 1900, 1901, 1902, 1903, 1~
## $ iso_code                                       <chr> "GBR", "GBR", "GBR", "GBR~
## $ population                                     <dbl> 41089602, 41459153, 41827~
## $ gdp                                            <dbl> 312531070000, 31219960800~
## $ biofuel_cons_change_pct                        <dbl> NA, NA, NA, NA, NA, NA, N~
```

```
## $ biofuel_cons_change_twh                    <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_cons_per_capita                    <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_consumption                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_elec_per_capita                    <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_electricity                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_share_elec                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ biofuel_share_energy                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ carbon_intensity_elec                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_cons_change_pct                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_cons_change_twh                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_cons_per_capita                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_consumption                           <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_elec_per_capita                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_electricity                           <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_prod_change_pct                       <dbl> NA, -2.724, 3.674, 1.426,~
## $ coal_prod_change_twh                       <dbl> NA, -42.668, 55.984, 22.5~
## $ coal_prod_per_capita                       <dbl> 38122.94, 36753.98, 37769~
## $ coal_production                            <dbl> 1566.457, 1523.789, 1579.~
## $ coal_share_elec                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ coal_share_energy                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ electricity_demand                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ electricity_demand_per_capita              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ electricity_generation                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ electricity_share_energy                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_cons_change_pct                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_cons_change_twh                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_per_capita                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ energy_per_gdp                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_cons_change_pct                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_cons_change_twh                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_elec_per_capita                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_electricity                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_energy_per_capita                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_fuel_consumption                    <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_share_elec                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ fossil_share_energy                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_cons_change_pct                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_cons_change_twh                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_consumption                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_elec_per_capita                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_electricity                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_energy_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_prod_change_pct                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_prod_change_twh                        <dbl> NA, 0, 0, 0, 0, 0, 0, 0, ~
## $ gas_prod_per_capita                        <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ gas_production                             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ gas_share_elec                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ gas_share_energy                           <dbl> NA, NA, NA, NA, NA, NA, N~
## $ greenhouse_gas_emissions                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_cons_change_pct                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_cons_change_twh                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_consumption                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_elec_per_capita                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_electricity                          <dbl> NA, NA, NA, NA, NA, NA, N~
```

```
## $ hydro_energy_per_capita                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_share_elec                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ hydro_share_energy                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_cons_change_pct                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_cons_change_twh                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_consumption                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_elec_per_capita                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_electricity                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_energy_per_capita                <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_share_elec                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ low_carbon_share_energy                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ net_elec_imports                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ net_elec_imports_share_demand               <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_cons_change_pct                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_cons_change_twh                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_consumption                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_elec_per_capita                     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_electricity                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_energy_per_capita                   <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_share_elec                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ nuclear_share_energy                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_cons_change_pct                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_cons_change_twh                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_consumption                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_elec_per_capita                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_electricity                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_energy_per_capita                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_prod_change_pct                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_prod_change_twh                         <dbl> NA, 0, 0, 0, 0, 0, 0, 0, ~
## $ oil_prod_per_capita                         <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ oil_production                              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ oil_share_elec                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ oil_share_energy                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewable_consumption                 <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewable_electricity                 <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewable_exc_biofuel_electricity     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_cons_change_pct            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_cons_change_twh            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_elec_per_capita            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_elec_per_capita_exc_biofuel <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_energy_per_capita          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_share_elec                 <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_share_elec_exc_biofuel     <dbl> NA, NA, NA, NA, NA, NA, N~
## $ other_renewables_share_energy               <dbl> NA, NA, NA, NA, NA, NA, N~
## $ per_capita_electricity                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ primary_energy_consumption                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_cons_change_pct                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_cons_change_twh                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_consumption                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_elec_per_capita                  <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_electricity                      <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_energy_per_capita                <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_share_elec                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ renewables_share_energy                     <dbl> NA, NA, NA, NA, NA, NA, N~
```

```
## $ solar_cons_change_pct                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_cons_change_twh                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_consumption                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_elec_per_capita                         <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_electricity                             <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_energy_per_capita                       <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_share_elec                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ solar_share_energy                            <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_cons_change_pct                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_cons_change_twh                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_consumption                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_elec_per_capita                          <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_electricity                              <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_energy_per_capita                        <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_share_elec                               <dbl> NA, NA, NA, NA, NA, NA, N~
## $ wind_share_energy                             <dbl> NA, NA, NA, NA, NA, NA, N~
```

Once we filter to UK our data set now has 130 columns and 126 rows.

```r
# Summary of UK data
summary(uk_data)
```

```
##    country               year        iso_code           population
##  Length:126         Min.   :1900   Length:126         Min.   :41089602
##  Class :character   1st Qu.:1931   Class :character   1st Qu.:46551092
##  Mode  :character   Median :1962   Mode  :character   Median :53462888
##                     Mean   :1962                      Mean   :53437750
##                     3rd Qu.:1994                      3rd Qu.:57967659
##                     Max.   :2025                      Max.   :69551334
##
##       gdp             biofuel_cons_change_pct biofuel_cons_change_twh
##  Min.   :2.993e+11   Min.   : -93.625        Min.   :-8.6510
##  1st Qu.:4.013e+11   1st Qu.:  -5.504        1st Qu.: 0.0000
##  Median :7.455e+11   Median :  15.066        Median : 0.0000
##  Mean   :1.033e+12   Mean   : 141.184        Mean   : 0.4695
##  3rd Qu.:1.503e+12   3rd Qu.:  84.585        3rd Qu.: 0.0090
##  Max.   :2.624e+12   Max.   :1358.741        Max.   :11.0360
##  NA's   :3           NA's   :104             NA's   :67
##  biofuel_cons_per_capita biofuel_consumption biofuel_elec_per_capita
##  Min.   :  0.00          Min.   : 0.000      Min.   : 10.46
##  1st Qu.:  0.00          1st Qu.: 0.000      1st Qu.: 55.02
##  Median :  0.00          Median : 0.000      Median :153.79
##  Mean   : 56.47          Mean   : 3.768      Mean   :239.19
##  3rd Qu.: 18.48          3rd Qu.: 1.148      3rd Qu.:462.44
##  Max.   :400.68          Max.   :27.702      Max.   :592.51
##  NA's   :66              NA's   :66          NA's   :90
##  biofuel_electricity biofuel_share_elec biofuel_share_energy
##  Min.   : 0.600      Min.   : 0.1880    Min.   :0.00000
##  1st Qu.: 3.235      1st Qu.: 0.8818    1st Qu.:0.00000
##  Median : 9.485      Median : 2.4150    Median :0.00000
##  Mean   :15.733      Mean   : 4.7744    Mean   :0.17910
##  3rd Qu.:30.527      3rd Qu.: 9.0068    3rd Qu.:0.04625
##  Max.   :41.210      Max.   :14.1080    Max.   :1.44500
```

```
##  NA's   :90          NA's   :90         NA's   :66
##  carbon_intensity_elec coal_cons_change_pct coal_cons_change_twh
##  Min.   :216.5         Min.   :-49.095      Min.   :-223.947
##  1st Qu.:325.6         1st Qu.: -9.841      1st Qu.: -52.566
##  Median :514.6         Median : -4.580      Median : -17.811
##  Mean   :467.1         Mean   : -4.735      Mean   : -22.345
##  3rd Qu.:554.8         3rd Qu.:  0.873      3rd Qu.:   3.919
##  Max.   :705.0         Max.   : 33.004      Max.   : 181.463
##  NA's   :90            NA's   :67           NA's   :67
##  coal_cons_per_capita coal_consumption  coal_elec_per_capita coal_electricity
##  Min.   :  679.3      Min.   :  46.97   Min.   :   4.745     Min.   :  0.33
##  1st Qu.: 6771.1      1st Qu.: 411.21   1st Qu.:1160.562     1st Qu.: 75.88
##  Median : 9615.1      Median : 550.87   Median :2087.309     Median :124.38
##  Mean   :10422.5      Mean   : 598.52   Mean   :1908.368     Mean   :113.39
##  3rd Qu.:14689.2      3rd Qu.: 826.53   3rd Qu.:2668.919     3rd Qu.:155.21
##  Max.   :25116.8      Max.   :1365.33   Max.   :3690.256     Max.   :211.46
##  NA's   :66           NA's   :66        NA's   :85           NA's   :85
##  coal_prod_change_pct coal_prod_change_twh coal_prod_per_capita
##  Min.   :-78.947      Min.   :-826.07      Min.   :   12.04
##  1st Qu.: -7.681      1st Qu.: -49.63      1st Qu.: 6549.68
##  Median : -2.825      Median : -10.87      Median :24938.26
##  Mean   : -3.789      Mean   : -12.63      Mean   :21522.01
##  3rd Qu.:  1.468      3rd Qu.:  14.18      3rd Qu.:33603.70
##  Max.   :100.075      Max.   : 874.17      Max.   :44037.05
##  NA's   :2            NA's   :2            NA's   :1
##  coal_production    coal_share_elec  coal_share_energy electricity_demand
##  Min.   :   0.832   Min.   : 0.113   Min.   : 2.45     Min.   :315.8
##  1st Qu.: 380.894   1st Qu.:22.392   1st Qu.:15.78     1st Qu.:339.7
##  Median :1304.267   Median :33.799   Median :22.24     Median :363.6
##  Mean   :1052.065   Mean   :32.944   Mean   :24.32     Mean   :363.6
##  3rd Qu.:1575.569   3rd Qu.:45.999   3rd Qu.:33.85     3rd Qu.:386.3
##  Max.   :1999.489   Max.   :69.485   Max.   :58.99     Max.   :406.7
##  NA's   :1          NA's   :85       NA's   :66        NA's   :90
##  electricity_demand_per_capita electricity_generation electricity_share_energy
##  Min.   :4590                  Min.   :284.2          Min.   :12.87
##  1st Qu.:5392                  1st Qu.:319.8          1st Qu.:14.01
##  Median :5893                  Median :338.2          Median :15.81
##  Mean   :5856                  Mean   :344.1          Mean   :15.60
##  3rd Qu.:6461                  3rd Qu.:376.8          3rd Qu.:16.91
##  Max.   :6711                  Max.   :398.4          Max.   :19.09
##  NA's   :90                    NA's   :85             NA's   :86
##  energy_cons_change_pct energy_cons_change_twh energy_per_capita
##  Min.   :-9.5820        Min.   :-232.913       Min.   :28016
##  1st Qu.:-1.5790        1st Qu.: -39.861       1st Qu.:39153
##  Median : 0.4040        Median :   8.722       Median :43332
##  Mean   :-0.2559        Mean   :  -6.428       Mean   :41101
##  3rd Qu.: 1.3875        3rd Qu.:  34.800       3rd Qu.:44639
##  Max.   : 4.9740        Max.   : 126.385       Max.   :47466
##  NA's   :67             NA's   :67             NA's   :66
##  energy_per_gdp  fossil_cons_change_pct fossil_cons_change_twh
##  Min.   :0.769   Min.   :-13.5580       Min.   :-229.69
##  1st Qu.:1.139   1st Qu.: -1.9165       1st Qu.: -44.19
##  Median :1.634   Median : -0.3570       Median :  -7.66
##  Mean   :1.663   Mean   : -0.7307       Mean   : -14.41
```

```
##  3rd Qu.:2.143   3rd Qu.:  1.5230       3rd Qu.:  35.05
##  Max.   :2.742   Max.   :  5.2680       Max.   : 128.93
##  NA's   :68      NA's   :67             NA's   :67
##  fossil_elec_per_capita fossil_electricity fossil_energy_per_capita
##  Min.   :1441           Min.   : 99.65     Min.   :20390
##  1st Qu.:2832           1st Qu.:185.17     1st Qu.:35209
##  Median :4148           Median :240.72     Median :39841
##  Mean   :3778           Mean   :229.79     Mean   :37093
##  3rd Qu.:4518           3rd Qu.:282.09     3rd Qu.:40785
##  Max.   :5104           Max.   :314.53     Max.   :45851
##  NA's   :85             NA's   :85         NA's   :66
##  fossil_fuel_consumption fossil_share_elec fossil_share_energy
##  Min.   :1410            Min.   :35.07     Min.   :73.54
##  1st Qu.:2136            1st Qu.:54.64     1st Qu.:88.77
##  Median :2287            Median :71.66     Median :90.65
##  Mean   :2188            Mean   :66.07     Mean   :89.80
##  3rd Qu.:2372            3rd Qu.:76.10     3rd Qu.:95.11
##  Max.   :2576            Max.   :80.86     Max.   :97.63
##  NA's   :66              NA's   :85        NA's   :66
##  gas_cons_change_pct gas_cons_change_twh gas_consumption    gas_elec_per_capita
##  Min.   :-16.828     Min.   :-165.743    Min.   :   8.385   Min.   :  30.71
##  1st Qu.: -1.272     1st Qu.:  -7.739    1st Qu.: 469.817   1st Qu.:1096.04
##  Median :  2.705     Median :  15.421    Median : 654.235   Median :1815.31
##  Mean   :  9.734     Mean   :  10.339    Mean   : 620.296   Mean   :1596.76
##  3rd Qu.:  7.085     3rd Qu.:  34.723    3rd Qu.: 829.349   3rd Qu.:2393.57
##  Max.   :125.804     Max.   : 121.754    Max.   :1019.879   Max.   :2841.01
##  NA's   :67          NA's   :67          NA's   :66         NA's   :85
##  gas_electricity   gas_energy_per_capita gas_prod_change_pct
##  Min.   :  1.743   Min.   :  153.4       Min.   :-100.000
##  1st Qu.: 63.740   1st Qu.: 8354.6       1st Qu.:  -3.652
##  Median :111.890   Median :10318.9       Median :   2.649
##  Mean   :100.034   Mean   :10246.3       Mean   :  13.041
##  3rd Qu.:143.130   3rd Qu.:13322.6       3rd Qu.:  11.457
##  Max.   :176.220   Max.   :17159.9       Max.   : 253.261
##  NA's   :85        NA's   :66            NA's   :55
##  gas_prod_change_twh gas_prod_per_capita gas_production     gas_share_elec
##  Min.   :-118.184    Min.   :    0.00    Min.   :   0.000   Min.   : 0.579
##  1st Qu.:   0.000    1st Qu.:    0.00    1st Qu.:   0.000   1st Qu.:18.890
##  Median :   0.000    Median :   24.07    Median :   1.282   Median :34.752
##  Mean   :   2.477    Mean   : 3988.23    Mean   : 238.475   Mean   :28.211
##  3rd Qu.:   3.910    3rd Qu.: 6745.88    3rd Qu.: 396.129   3rd Qu.:39.397
##  Max.   : 140.154    Max.   :19211.68    Max.   :1134.590   Max.   :45.974
##  NA's   :2           NA's   :1           NA's   :1          NA's   :85
##  gas_share_energy greenhouse_gas_emissions hydro_cons_change_pct
##  Min.   : 0.361   Min.   : 61.52           Min.   :-32.580
##  1st Qu.:19.285   1st Qu.: 95.34           1st Qu.:-10.227
##  Median :28.134   Median :183.85           Median :  1.619
##  Mean   :25.612   Mean   :156.31           Mean   :  1.909
##  3rd Qu.:35.508   3rd Qu.:203.54           3rd Qu.: 11.148
##  Max.   :39.904   Max.   :222.21           Max.   : 58.484
##  NA's   :66       NA's   :101              NA's   :67
##  hydro_cons_change_twh hydro_consumption hydro_elec_per_capita
##  Min.   :-4.36900      Min.   : 8.795    Min.   : 53.99
##  1st Qu.:-1.29300      1st Qu.:11.550    1st Qu.: 72.99
```

```
##  Median : 0.10500      Median :12.867    Median : 80.58
##  Mean   : 0.02117      Mean   :12.666    Mean   : 78.84
##  3rd Qu.: 1.25700      3rd Qu.:13.693    3rd Qu.: 84.02
##  Max.   : 5.39300      Max.   :16.983    Max.   :102.15
##  NA's   :67            NA's   :66        NA's   :65
##  hydro_electricity hydro_energy_per_capita hydro_share_elec hydro_share_energy
##  Min.   :3.230     Min.   :147.0           Min.   :0.811    Min.   :0.3300
##  1st Qu.:4.187     1st Qu.:197.1           1st Qu.:1.312    1st Qu.:0.4600
##  Median :4.700     Median :218.3           Median :1.450    Median :0.5310
##  Mean   :4.732     Mean   :212.3           Mean   :1.479    Mean   :0.5284
##  3rd Qu.:5.310     3rd Qu.:230.6           3rd Qu.:1.666    3rd Qu.:0.5705
##  Max.   :6.880     Max.   :257.5           Max.   :2.217    Max.   :0.8650
##  NA's   :65        NA's   :66              NA's   :85       NA's   :66
##  low_carbon_cons_change_pct low_carbon_cons_change_twh low_carbon_consumption
##  Min.   :-9.328             Min.   :-39.762            Min.   : 54.85
##  1st Qu.:-2.322             1st Qu.: -4.141            1st Qu.:118.21
##  Median : 4.873             Median :  8.011            Median :234.08
##  Mean   : 4.445             Mean   :  7.666            Mean   :239.22
##  3rd Qu.:10.347             3rd Qu.: 18.124            3rd Qu.:292.44
##  Max.   :25.351             Max.   : 64.262            Max.   :507.14
##  NA's   :67                 NA's   :67                 NA's   :66
##  low_carbon_elec_per_capita low_carbon_electricity low_carbon_energy_per_capita
##  Min.   : 363.3             Min.   : 19.75         Min.   :1009
##  1st Qu.: 758.0             1st Qu.: 42.61         1st Qu.:2102
##  Median :1440.1             Median : 88.32         Median :3883
##  Mean   :1437.5             Mean   : 88.76         Mean   :3893
##  3rd Qu.:1780.3             3rd Qu.:105.33         3rd Qu.:4836
##  Max.   :2730.0             Max.   :188.77         Max.   :7399
##  NA's   :65                 NA's   :65             NA's   :66
##  low_carbon_share_elec low_carbon_share_energy net_elec_imports
##  Min.   :19.14         Min.   : 2.370          Min.   :-4.300
##  1st Qu.:23.90         1st Qu.: 4.891          1st Qu.: 8.387
##  Median :28.34         Median : 9.351          Median :14.595
##  Mean   :33.79         Mean   :10.197          Mean   :14.073
##  3rd Qu.:45.36         3rd Qu.:11.234          3rd Qu.:17.875
##  Max.   :64.93         Max.   :26.457          Max.   :33.190
##  NA's   :85            NA's   :66              NA's   :90
##  net_elec_imports_share_demand nuclear_cons_change_pct nuclear_cons_change_twh
##  Min.   :-1.343                Min.   :-16.727         Min.   :-34.1110
##  1st Qu.: 2.106                1st Qu.: -7.242         1st Qu.: -9.7995
##  Median : 4.027                Median :  1.963         Median :  2.6040
##  Mean   : 3.990                Mean   :  2.269         Mean   :  0.9653
##  3rd Qu.: 5.084                3rd Qu.: 10.521         3rd Qu.: 12.9530
##  Max.   :10.458                Max.   : 33.578         Max.   : 42.8740
##  NA's   :90                    NA's   :67              NA's   :67
##  nuclear_consumption nuclear_elec_per_capita nuclear_electricity
##  Min.   : 42.04      Min.   : 278.4          Min.   :15.13
##  1st Qu.:103.26      1st Qu.: 657.8          1st Qu.:37.97
##  Median :162.03      Median : 982.8          Median :61.09
##  Mean   :157.65      Mean   : 969.5          Mean   :58.12
##  3rd Qu.:199.88      3rd Qu.:1235.8          3rd Qu.:71.73
##  Max.   :276.35      Max.   :1697.0          Max.   :99.49
##  NA's   :66          NA's   :65              NA's   :65
##  nuclear_energy_per_capita nuclear_share_elec nuclear_share_energy
```

```
##  Min.   : 773.4           Min.   :12.43      Min.   : 1.816
##  1st Qu.:1767.3           1st Qu.:18.28      1st Qu.: 4.443
##  Median :2653.1           Median :20.49      Median : 6.623
##  Mean   :2640.0           Mean   :20.35      Mean   : 6.442
##  3rd Qu.:3424.6           3rd Qu.:22.69      3rd Qu.: 8.036
##  Max.   :4713.7           Max.   :27.99      Max.   :10.450
##  NA's   :66               NA's   :85         NA's   :66
##  oil_cons_change_pct oil_cons_change_twh oil_consumption  oil_elec_per_capita
##  Min.   :-21.2030    Min.   :-179.476    Min.   : 667.0   Min.   :129.8
##  1st Qu.: -1.8015    1st Qu.: -16.659    1st Qu.: 885.3   1st Qu.:161.5
##  Median :  0.2290    Median :   2.130    Median : 963.2   Median :201.0
##  Mean   : -0.1093    Mean   :  -2.399    Mean   : 968.8   Mean   :272.9
##  3rd Qu.:  2.3570    3rd Qu.:  22.683    3rd Qu.:1012.8   3rd Qu.:328.5
##  Max.   : 22.9240    Max.   : 200.071    Max.   :1344.5   Max.   :853.4
##  NA's   :67          NA's   :67          NA's   :66       NA's   :85
##  oil_electricity oil_energy_per_capita oil_prod_change_pct oil_prod_change_twh
##  Min.   : 8.36   Min.   : 9903         Min.   :-33.858     Min.   :-264.5710
##  1st Qu.:10.64   1st Qu.:14377         1st Qu.: -7.209     1st Qu.:  -0.2037
##  Median :12.41   Median :16408         Median :  1.190     Median :   0.0000
##  Mean   :16.37   Mean   :16425         Mean   : 23.481     Mean   :   2.8511
##  3rd Qu.:19.15   3rd Qu.:17461         3rd Qu.: 11.375     3rd Qu.:   0.1540
##  Max.   :48.31   Max.   :23928         Max.   :676.084     Max.   : 306.4850
##  NA's   :85      NA's   :66            NA's   :41          NA's   :2
##  oil_prod_per_capita oil_production   oil_share_elec   oil_share_energy
##  Min.   :    0.00    Min.   :   0.0   Min.   : 2.333   Min.   :33.98
##  1st Qu.:    0.00    1st Qu.:   0.0   1st Qu.: 3.024   1st Qu.:37.08
##  Median :   19.04    Median :   1.0   Median : 3.534   Median :38.86
##  Mean   : 6372.02    Mean   : 377.6   Mean   : 4.920   Mean   :39.87
##  3rd Qu.:11621.72    3rd Qu.: 732.2   3rd Qu.: 5.458   3rd Qu.:40.76
##  Max.   :27098.87    Max.   :1594.5   Max.   :16.206   Max.   :51.60
##  NA's   :1           NA's   :1        NA's   :85       NA's   :66
##  other_renewable_consumption other_renewable_electricity
##  Min.   :  0.000            Min.   : 0.000
##  1st Qu.:  0.000            1st Qu.: 0.000
##  Median :  4.938            Median : 1.640
##  Mean   : 27.355            Mean   : 9.286
##  3rd Qu.: 34.694            3rd Qu.:12.260
##  Max.   :125.295            Max.   :41.210
##  NA's   :66                 NA's   :65
##  other_renewable_exc_biofuel_electricity other_renewables_cons_change_pct
##  Min.   :0.000000                        Min.   :-10.385
##  1st Qu.:0.000000                        1st Qu.:  6.302
##  Median :0.000000                        Median : 12.723
##  Mean   :0.000667                        Mean   : 13.679
##  3rd Qu.:0.000000                        3rd Qu.: 22.515
##  Max.   :0.010000                        Max.   : 35.805
##  NA's   :96                              NA's   :92
##  other_renewables_cons_change_twh other_renewables_elec_per_capita
##  Min.   :-12.983                  Min.   :  0.0
##  1st Qu.:  0.000                  1st Qu.:  0.0
##  Median :  0.509                  Median : 28.2
##  Mean   :  2.124                  Mean   :141.2
##  3rd Qu.:  3.261                  3rd Qu.:194.6
##  Max.   : 20.743                  Max.   :592.5
```

```
##  NA's   :67                       NA's   :65
##  other_renewables_elec_per_capita_exc_biofuel
##  Min.   :0.000000
##  1st Qu.:0.000000
##  Median :0.000000
##  Mean   :0.009967
##  3rd Qu.:0.000000
##  Max.   :0.150000
##  NA's   :96
##  other_renewables_energy_per_capita other_renewables_share_elec
##  Min.   :   0.00                    Min.   : 0.000
##  1st Qu.:   0.00                    1st Qu.: 0.486
##  Median :  85.01                    Median : 2.284
##  Mean   : 417.63                    Mean   : 4.192
##  3rd Qu.: 553.83                    3rd Qu.: 8.635
##  Max.   :1847.53                    Max.   :14.108
##  NA's   :66                         NA's   :85
##  other_renewables_share_elec_exc_biofuel other_renewables_share_energy
##  Min.   :0e+00                           Min.   :0.000
##  1st Qu.:0e+00                           1st Qu.:0.000
##  Median :0e+00                           Median :0.194
##  Mean   :2e-04                           Mean   :1.251
##  3rd Qu.:0e+00                           3rd Qu.:1.428
##  Max.   :3e-03                           Max.   :6.537
##  NA's   :96                              NA's   :66
##  per_capita_electricity primary_energy_consumption renewables_cons_change_pct
##  Min.   :4110           Min.   :1930               Min.   :-25.639
##  1st Qu.:5183           1st Qu.:2316               1st Qu.: -3.041
##  Median :5584           Median :2479               Median :  6.095
##  Mean   :5599           Mean   :2434               Mean   :  7.482
##  3rd Qu.:6187           3rd Qu.:2565               3rd Qu.: 17.852
##  Max.   :6657           Max.   :2713               Max.   : 54.688
##  NA's   :85             NA's   :66                 NA's   :67
##  renewables_cons_change_twh renewables_consumption renewables_elec_per_capita
##  Min.   :-29.2710           Min.   :  9.044        Min.   :  58.71
##  1st Qu.: -0.6595           1st Qu.: 12.600        1st Qu.:  80.52
##  Median :  1.2920           Median : 18.506        Median : 114.87
##  Mean   :  6.7006           Mean   : 81.568        Mean   : 467.98
##  3rd Qu.:  5.2465           3rd Qu.: 72.934        3rd Qu.: 415.51
##  Max.   : 48.3820           Max.   :408.147        Max.   :2191.04
##  NA's   :67                 NA's   :66             NA's   :65
##  renewables_electricity renewables_energy_per_capita renewables_share_elec
##  Min.   :  3.256        Min.   : 163.1               Min.   : 1.327
##  1st Qu.:  4.536        1st Qu.: 223.6               1st Qu.: 1.980
##  Median :  6.680        Median : 319.4               Median : 4.250
##  Mean   : 30.642        Mean   :1252.6               Mean   :13.435
##  3rd Qu.: 26.180        3rd Qu.:1165.4               3rd Qu.:24.474
##  Max.   :152.390        Max.   :5903.4               Max.   :52.049
##  NA's   :65             NA's   :66                   NA's   :85
##  renewables_share_energy solar_cons_change_pct solar_cons_change_twh
##  Min.   : 0.3620         Min.   :-100.00       Min.   :-1.1070
##  1st Qu.: 0.5278         1st Qu.:   0.00       1st Qu.: 0.0000
##  Median : 0.7280         Median :  17.65       Median : 0.0000
##  Mean   : 3.7548         Mean   :  53.80       Mean   : 0.6114
```

```
##  3rd Qu.: 3.0025        3rd Qu.:  50.00      3rd Qu.: 0.0080  
##  Max.   :21.2930        Max.   : 504.97      Max.   : 8.7550  
##  NA's   :66             NA's   :97           NA's   :67       
##  solar_consumption  solar_elec_per_capita solar_electricity
##  Min.   : 0.00000   Min.   :  0.000       Min.   : 0.000   
##  1st Qu.: 0.00000   1st Qu.:  0.000       1st Qu.: 0.000   
##  Median : 0.00150   Median :  0.000       Median : 0.000   
##  Mean   : 5.32662   Mean   : 36.002       Mean   : 2.431   
##  3rd Qu.: 0.06525   3rd Qu.:  0.635       3rd Qu.: 0.040   
##  Max.   :36.07200   Max.   :277.780       Max.   :19.320   
##  NA's   :66         NA's   :65            NA's   :65       
##  solar_energy_per_capita solar_share_elec solar_share_energy
##  Min.   :  0.0000        Min.   :0.000    Min.   :0.0000    
##  1st Qu.:  0.0000        1st Qu.:0.000    1st Qu.:0.0000    
##  Median :  0.0235        Median :0.003    Median :0.0000    
##  Mean   : 79.2686        Mean   :1.150    Mean   :0.2563    
##  3rd Qu.:  1.0462        3rd Qu.:2.222    3rd Qu.:0.0025    
##  Max.   :521.7370        Max.   :6.599    Max.   :1.8820    
##  NA's   :66              NA's   :85       NA's   :66        
##  wind_cons_change_pct wind_cons_change_twh wind_consumption
##  Min.   :-14.148      Min.   :-26.813      Min.   :  0.000 
##  1st Qu.:  6.564      1st Qu.:  0.000      1st Qu.:  0.000 
##  Median : 24.332      Median :  0.038      Median :  1.022 
##  Mean   : 43.593      Mean   :  3.475      Mean   : 32.452 
##  3rd Qu.: 35.627      3rd Qu.:  2.780      3rd Qu.: 24.966 
##  Max.   :557.576      Max.   : 37.982      Max.   :205.019 
##  NA's   :91           NA's   :67           NA's   :66      
##  wind_elec_per_capita wind_electricity wind_energy_per_capita wind_share_elec
##  Min.   :   0.000     Min.   : 0.00    Min.   :   0.0         Min.   : 0.000 
##  1st Qu.:   0.000     1st Qu.: 0.00    1st Qu.:   0.0         1st Qu.: 0.116 
##  Median :   6.706     Median : 0.39    Median :  17.6         Median : 0.728 
##  Mean   : 211.973     Mean   :14.19    Mean   : 487.0         Mean   : 6.613 
##  3rd Qu.: 163.317     3rd Qu.:10.29    3rd Qu.: 398.6         3rd Qu.:10.956 
##  Max.   :1240.954     Max.   :86.31    Max.   :2965.4         Max.   :29.479 
##  NA's   :65           NA's   :65       NA's   :66             NA's   :85     
##  wind_share_energy
##  Min.   : 0.0000  
##  1st Qu.: 0.0000  
##  Median : 0.0405  
##  Mean   : 1.5400  
##  3rd Qu.: 1.0272  
##  Max.   :10.6960  
##  NA's   :66
```

```r
# Total number of missing values
sum(is.na(uk_data))
```

```
## [1] 8437
```

The above code chunk has revealed the type of variables that make up the uk_data, the missing values and the summary statistics of each numerical variable. It was also observed that the missing values have now reduced to 8437. Based on the uk_data most of the data before 2000 is missing and we will not be able to make a comprehensive analysis, therefore we will drop the data before 2000. In the code chunk below we filter our data to the years 2000 onwards.

```r
# Filter to years 2000 onwards
uk_energy <- uk_data %>%
  filter(year >= 2000)
# Glimpse of uk energy
glimpse(uk_energy)
```

```
## Rows: 26
## Columns: 130
## $ country                                 <chr> "United Kingdom", "United~
## $ year                                    <dbl> 2000, 2001, 2002, 2003, 2~
## $ iso_code                                <chr> "GBR", "GBR", "GBR", "GBR~
## $ population                              <dbl> 59057333, 59288095, 59540~
## $ gdp                                     <dbl> 1.889565e+12, 1.939268e+1~
## $ biofuel_cons_change_pct                 <dbl> NA, NA, NA, 950.000, -9.5~
## $ biofuel_cons_change_twh                 <dbl> 0.000, 0.000, 0.018, 0.17~
## $ biofuel_cons_per_capita                 <dbl> 0.000, 0.000, 0.304, 3.17~
## $ biofuel_consumption                     <dbl> 0.000, 0.000, 0.018, 0.19~
## $ biofuel_elec_per_capita                 <dbl> 65.699, 76.407, 85.320, 1~
## $ biofuel_electricity                     <dbl> 3.88, 4.53, 5.08, 6.17, 7~
## $ biofuel_share_elec                      <dbl> 1.029, 1.177, 1.312, 1.54~
## $ biofuel_share_energy                    <dbl> 0.000, 0.000, 0.001, 0.00~
## $ carbon_intensity_elec                   <dbl> 521.99, 528.99, 520.89, 5~
## $ coal_cons_change_pct                    <dbl> 7.077, 5.946, -8.228, 6.7~
## $ coal_cons_change_twh                    <dbl> 28.231, 25.398, -37.237, ~
## $ coal_cons_per_capita                    <dbl> 7232.765, 7632.989, 6975.~
## $ coal_consumption                        <dbl> 427.148, 452.545, 415.308~
## $ coal_elec_per_capita                    <dbl> 2031.077, 2217.309, 2087.~
## $ coal_electricity                        <dbl> 119.95, 131.46, 124.28, 1~
## $ coal_prod_change_pct                    <dbl> -15.796, 2.134, -5.815, -~
## $ coal_prod_change_twh                    <dbl> -42.656, 4.853, -13.505, ~
## $ coal_prod_per_capita                    <dbl> 3850.219, 3917.089, 3673.~
## $ coal_production                         <dbl> 227.384, 232.237, 218.732~
## $ coal_share_elec                         <dbl> 31.812, 34.166, 32.094, 3~
## $ coal_share_energy                       <dbl> 16.072, 16.851, 15.827, 1~
## $ electricity_demand                      <dbl> 391.23, 395.17, 395.65, 4~
## $ electricity_demand_per_capita           <dbl> 6624.580, 6665.250, 6645.~
## $ electricity_generation                  <dbl> 377.06, 384.77, 387.24, 3~
## $ electricity_share_energy                <dbl> 15.158, 15.340, 15.806, 1~
## $ energy_cons_change_pct                  <dbl> 0.943, 1.039, -2.265, 1.4~
## $ energy_cons_change_twh                  <dbl> 24.897, 27.690, -60.966, ~
## $ energy_per_capita                       <dbl> 45110.68, 45402.14, 44185~
## $ energy_per_gdp                          <dbl> 1.410, 1.388, 1.326, 1.30~
## $ fossil_cons_change_pct                  <dbl> 2.243, 0.673, -2.397, 1.6~
## $ fossil_cons_change_twh                  <dbl> 52.481, 16.107, -57.733, ~
## $ fossil_elec_per_capita                  <dbl> 4776.545, 4809.397, 4841.~
## $ fossil_electricity                      <dbl> 282.09, 285.14, 288.27, 2~
## $ fossil_energy_per_capita                <dbl> 40511.81, 40625.80, 39483~
## $ fossil_fuel_consumption                 <dbl> 2392.519, 2408.626, 2350.~
## $ fossil_share_elec                       <dbl> 74.813, 74.107, 74.442, 7~
## $ fossil_share_energy                     <dbl> 90.022, 89.689, 89.591, 8~
## $ gas_cons_change_pct                     <dbl> 3.529, -0.482, -1.308, 0.~
## $ gas_cons_change_twh                     <dbl> 34.540, -4.885, -13.190, ~
## $ gas_consumption                         <dbl> 1013.417, 1008.532, 995.3~
```

```
## $ gas_elec_per_capita                          <dbl> 2507.394, 2393.567, 2557.~
## $ gas_electricity                              <dbl> 148.08, 141.91, 152.28, 1~
## $ gas_energy_per_capita                        <dbl> 17159.887, 17010.699, 167~
## $ gas_prod_change_pct                          <dbl> 9.372, -2.332, -2.100, -0~
## $ gas_prod_change_twh                          <dbl> 97.224, -26.454, -23.272,~
## $ gas_prod_per_capita                          <dbl> 19211.678, 18690.713, 182~
## $ gas_production                               <dbl> 1134.590, 1108.137, 1084.~
## $ gas_share_elec                               <dbl> 39.272, 36.882, 39.324, 3~
## $ gas_share_energy                             <dbl> 38.131, 37.554, 37.932, 3~
## $ greenhouse_gas_emissions                     <dbl> 196.82, 203.54, 201.71, 2~
## $ hydro_cons_change_pct                        <dbl> -4.683, -20.274, 18.074, ~
## $ hydro_cons_change_twh                        <dbl> -0.694, -2.926, 1.916, -4~
## $ hydro_consumption                            <dbl> 14.127, 11.201, 13.116, 8~
## $ hydro_elec_per_capita                        <dbl> 86.187, 68.311, 80.449, 5~
## $ hydro_electricity                            <dbl> 5.09, 4.05, 4.79, 3.23, 4~
## $ hydro_energy_per_capita                      <dbl> 239.209, 188.920, 220.293~
## $ hydro_share_elec                             <dbl> 1.350, 1.053, 1.237, 0.81~
## $ hydro_share_energy                           <dbl> 0.532, 0.417, 0.500, 0.33~
## $ low_carbon_cons_change_pct                   <dbl> -9.328, 4.910, -0.647, 0.~
## $ low_carbon_cons_change_twh                   <dbl> -26.985, 11.717, -3.743, ~
## $ low_carbon_consumption                       <dbl> 265.174, 276.891, 273.148~
## $ low_carbon_elec_per_capita                   <dbl> 1608.098, 1680.438, 1662.~
## $ low_carbon_electricity                       <dbl> 94.97, 99.63, 98.97, 99.3~
## $ low_carbon_energy_per_capita                 <dbl> 4490.106, 4670.255, 4587.~
## $ low_carbon_share_elec                        <dbl> 25.187, 25.893, 25.558, 2~
## $ low_carbon_share_energy                      <dbl> 9.978, 10.311, 10.409, 10~
## $ net_elec_imports                             <dbl> 14.17, 10.40, 8.41, 2.16,~
## $ net_elec_imports_share_demand                <dbl> 3.622, 2.632, 2.126, 0.54~
## $ nuclear_cons_change_pct                      <dbl> -10.585, 5.913, -2.491, 0~
## $ nuclear_cons_change_twh                      <dbl> -27.973, 12.589, -8.194, ~
## $ nuclear_consumption                          <dbl> 236.285, 248.875, 240.680~
## $ nuclear_elec_per_capita                      <dbl> 1440.295, 1519.529, 1475.~
## $ nuclear_electricity                          <dbl> 85.06, 90.09, 87.85, 88.6~
## $ nuclear_energy_per_capita                    <dbl> 4000.950, 4197.720, 4042.~
## $ nuclear_share_elec                           <dbl> 22.559, 23.414, 22.686, 2~
## $ nuclear_share_energy                         <dbl> 8.891, 9.267, 9.172, 9.07~
## $ oil_cons_change_pct                          <dbl> -1.069, -0.463, -0.771, 0~
## $ oil_cons_change_twh                          <dbl> -10.290, -4.405, -7.306, ~
## $ oil_consumption                              <dbl> 951.954, 947.549, 940.243~
## $ oil_elec_per_capita                          <dbl> 238.074, 198.522, 196.672~
## $ oil_electricity                              <dbl> 14.06, 11.77, 11.71, 11.4~
## $ oil_energy_per_capita                        <dbl> 16119.154, 15982.114, 157~
## $ oil_prod_change_pct                          <dbl> -7.917, -7.578, -0.629, -~
## $ oil_prod_change_twh                          <dbl> -126.232, -111.260, -8.53~
## $ oil_prod_per_capita                          <dbl> 24861.088, 22887.725, 226~
## $ oil_production                               <dbl> 1468.229, 1356.970, 1348.~
## $ oil_share_elec                               <dbl> 3.729, 3.059, 3.024, 2.88~
## $ oil_share_energy                             <dbl> 35.819, 35.284, 35.832, 3~
## $ other_renewable_consumption                  <dbl> 12.131, 14.144, 15.875, 1~
## $ other_renewable_electricity                  <dbl> 3.88, 4.53, 5.08, 6.17, 7~
## $ other_renewable_exc_biofuel_electricity      <dbl> 0.00, 0.00, 0.00, 0.00, 0~
## $ other_renewables_cons_change_pct             <dbl> 13.202, 16.595, 12.243, 2~
## $ other_renewables_cons_change_twh             <dbl> 1.415, 2.013, 1.732, 3.41~
## $ other_renewables_elec_per_capita             <dbl> 65.699, 76.407, 85.320, 1~
```

```
## $ other_renewables_elec_per_capita_exc_biofuel <dbl> 0.000, 0.000, 0.000, 0.00~
## $ other_renewables_energy_per_capita           <dbl> 205.404, 238.560, 266.629~
## $ other_renewables_share_elec                  <dbl> 1.029, 1.177, 1.312, 1.54~
## $ other_renewables_share_elec_exc_biofuel      <dbl> 0.000, 0.000, 0.000, 0.00~
## $ other_renewables_share_energy                <dbl> 0.456, 0.527, 0.605, 0.72~
## $ per_capita_electricity                       <dbl> 6384.643, 6489.835, 6503.~
## $ primary_energy_consumption                   <dbl> 2664.116, 2691.806, 2630.~
## $ renewables_cons_change_pct                   <dbl> 3.108, -3.701, 16.758, -2~
## $ renewables_cons_change_twh                   <dbl> 0.987, -0.873, 4.452, -0.~
## $ renewables_consumption                       <dbl> 28.888, 28.016, 32.467, 3~
## $ renewables_elec_per_capita                   <dbl> 167.803, 160.909, 186.763~
## $ renewables_electricity                       <dbl> 9.91, 9.54, 11.12, 10.69,~
## $ renewables_energy_per_capita                 <dbl> 489.156, 472.535, 545.296~
## $ renewables_share_elec                        <dbl> 2.628, 2.479, 2.872, 2.68~
## $ renewables_share_energy                      <dbl> 1.087, 1.043, 1.237, 1.19~
## $ solar_cons_change_pct                        <dbl> 0.000, 100.000, 50.000, 0~
## $ solar_cons_change_twh                        <dbl> 0.000, 0.003, 0.003, 0.00~
## $ solar_consumption                            <dbl> 0.003, 0.006, 0.008, 0.00~
## $ solar_elec_per_capita                        <dbl> 0.000, 0.000, 0.000, 0.00~
## $ solar_electricity                            <dbl> 0.00, 0.00, 0.00, 0.00, 0~
## $ solar_energy_per_capita                      <dbl> 0.047, 0.093, 0.138, 0.13~
## $ solar_share_elec                             <dbl> 0.000, 0.000, 0.000, 0.00~
## $ solar_share_energy                           <dbl> 0.000, 0.000, 0.000, 0.00~
## $ wind_cons_change_pct                         <dbl> 11.294, 2.008, 30.466, 2.~
## $ wind_cons_change_twh                         <dbl> 0.267, 0.038, 0.784, 0.06~
## $ wind_consumption                             <dbl> 2.628, 2.666, 3.449, 3.51~
## $ wind_elec_per_capita                         <dbl> 15.917, 16.192, 20.994, 2~
## $ wind_electricity                             <dbl> 0.94, 0.96, 1.25, 1.29, 1~
## $ wind_energy_per_capita                       <dbl> 44.495, 44.963, 57.932, 5~
## $ wind_share_elec                              <dbl> 0.249, 0.249, 0.323, 0.32~
## $ wind_share_energy                            <dbl> 0.099, 0.099, 0.131, 0.13~
```

From the code chunk above, once we filter to years 2000 onwards, our data set reduces to 130 columns and 126 rows.

```r
# Summary of UK data
summary(uk_energy)
```

```
##    country               year        iso_code           population
##  Length:26          Min.   :2000   Length:26          Min.   :59057333
##  Class :character   1st Qu.:2006   Class :character   1st Qu.:61172284
##  Mode  :character   Median :2012   Mode  :character   Median :64174937
##                     Mean   :2012                      Mean   :64130404
##                     3rd Qu.:2019                      3rd Qu.:67021337
##                     Max.   :2025                      Max.   :69551334
##
##       gdp             biofuel_cons_change_pct biofuel_cons_change_twh
##  Min.   :1.890e+12   Min.   : -93.625        Min.   :-8.651
##  1st Qu.:2.157e+12   1st Qu.:  -5.504        1st Qu.:-0.030
##  Median :2.255e+12   Median :  15.066        Median : 0.172
##  Mean   :2.281e+12   Mean   : 141.184        Mean   : 1.108
##  3rd Qu.:2.453e+12   3rd Qu.:  84.585        3rd Qu.: 1.963
##  Max.   :2.624e+12   Max.   :1358.741        Max.   :11.036
```

```
##  NA's   :3           NA's   :4              NA's   :1
##  biofuel_cons_per_capita biofuel_consumption biofuel_elec_per_capita
##  Min.   :  0.000         Min.   : 0.000      Min.   : 65.7
##  1st Qu.:  9.424         1st Qu.: 0.589      1st Qu.:151.6
##  Median :148.963         Median : 9.240      Median :255.8
##  Mean   :135.530         Mean   : 9.044      Mean   :320.3
##  3rd Qu.:228.605         3rd Qu.:15.260      3rd Qu.:518.5
##  Max.   :400.675         Max.   :27.702      Max.   :592.5
##  NA's   :1               NA's   :1
##  biofuel_electricity biofuel_share_elec biofuel_share_energy
##  Min.   : 3.88       Min.   : 1.029     Min.   :0.0000
##  1st Qu.: 9.29       1st Qu.: 2.339     1st Qu.:0.0240
##  Median :16.42       Median : 4.551     Median :0.3620
##  Mean   :21.15       Mean   : 6.426     Mean   :0.4298
##  3rd Qu.:34.86       3rd Qu.:10.914     3rd Qu.:0.6850
##  Max.   :41.21       Max.   :14.108     Max.   :1.4450
##                                         NA's   :1
##  carbon_intensity_elec coal_cons_change_pct coal_cons_change_twh
##  Min.   :216.5         Min.   :-49.095      Min.   :-136.151
##  1st Qu.:272.7         1st Qu.:-15.593      1st Qu.: -28.098
##  Median :502.1         Median : -7.189      Median :  -8.222
##  Mean   :419.2         Mean   : -6.939      Mean   : -14.078
##  3rd Qu.:533.7         3rd Qu.:  2.340      3rd Qu.:   7.958
##  Max.   :559.3         Max.   : 26.153      Max.   :  93.550
##                        NA's   :1            NA's   :1
##  coal_cons_per_capita coal_consumption coal_elec_per_capita coal_electricity
##  Min.   : 679.3       Min.   : 46.97   Min.   :   4.745     Min.   :  0.330
##  1st Qu.:1444.4       1st Qu.: 96.42   1st Qu.: 140.366     1st Qu.:  9.398
##  Median :5631.3       Median :354.81   Median :1677.972     Median :105.315
##  Mean   :4660.2       Mean   :289.77   Mean   :1279.212     Mean   : 79.207
##  3rd Qu.:7188.1       3rd Qu.:434.93   3rd Qu.:2164.548     3rd Qu.:131.160
##  Max.   :7878.5       Max.   :480.99   Max.   :2438.122     Max.   :148.850
##  NA's   :1            NA's   :1
##  coal_prod_change_pct coal_prod_change_twh coal_prod_per_capita
##  Min.   :-78.947      Min.   :-42.656      Min.   :  12.04
##  1st Qu.:-26.135      1st Qu.:-14.318      1st Qu.: 314.62
##  Median :-10.193      Median : -7.559      Median :1828.08
##  Mean   :-17.411      Mean   :-10.768      Mean   :1569.89
##  3rd Qu.: -6.113      3rd Qu.: -1.857      3rd Qu.:2066.32
##  Max.   :  5.690      Max.   :  6.724      Max.   :3917.09
##  NA's   :1            NA's   :1            NA's   :1
##  coal_production   coal_share_elec  coal_share_energy electricity_demand
##  Min.   :  0.832   Min.   : 0.113   Min.   : 2.450    Min.   :315.8
##  1st Qu.: 21.002   1st Qu.: 2.914   1st Qu.: 4.326    1st Qu.:349.5
##  Median :116.925   Median :28.815   Median :15.394    Median :373.4
##  Mean   : 96.484   Mean   :21.047   Mean   :11.702    Mean   :367.3
##  3rd Qu.:127.411   3rd Qu.:33.713   3rd Qu.:16.464    3rd Qu.:395.5
##  Max.   :232.237   Max.   :39.242   Max.   :19.072    Max.   :406.7
##  NA's   :1                          NA's   :1
##  electricity_demand_per_capita electricity_generation electricity_share_energy
##  Min.   :4590                  Min.   :284.2          Min.   :15.16
##  1st Qu.:5215                  1st Qu.:328.8          1st Qu.:16.00
##  Median :5831                  Median :361.1          Median :16.68
##  Mean   :5766                  Mean   :353.8          Mean   :16.75
```

```
##  3rd Qu.:6602                  3rd Qu.:386.6         3rd Qu.:17.25
##  Max.   :6711                  Max.   :398.4         Max.   :19.09
##                                                      NA's   :1
##  energy_cons_change_pct energy_cons_change_twh energy_per_capita
##  Min.   :-9.582         Min.   :-210.142       Min.   :28016
##  1st Qu.:-2.379         1st Qu.: -60.966       1st Qu.:33653
##  Median : 0.183         Median :   4.121       Median :37094
##  Mean   :-1.187         Mean   : -28.090       Mean   :37347
##  3rd Qu.: 0.943         3rd Qu.:  22.031       3rd Qu.:43870
##  Max.   : 2.160         Max.   :  52.335       Max.   :45402
##  NA's   :1              NA's   :1              NA's   :1
##  energy_per_gdp  fossil_cons_change_pct fossil_cons_change_twh
##  Min.   :0.769   Min.   :-13.558        Min.   :-229.69
##  1st Qu.:0.896   1st Qu.: -2.397        1st Qu.: -57.73
##  Median :1.053   Median : -1.410        Median : -27.27
##  Mean   :1.072   Mean   : -1.929        Mean   : -37.21
##  3rd Qu.:1.232   3rd Qu.:  0.673        3rd Qu.:  16.11
##  Max.   :1.410   Max.   :  3.443        Max.   :  69.05
##  NA's   :3       NA's   :1              NA's   :1
##  fossil_elec_per_capita fossil_electricity fossil_energy_per_capita
##  Min.   :1441           Min.   : 99.65     Min.   :20390
##  1st Qu.:2284           1st Qu.:153.09     1st Qu.:26297
##  Median :3792           Median :243.34     Median :32213
##  Mean   :3543           Mean   :223.04     Mean   :31749
##  3rd Qu.:4834           3rd Qu.:292.37     3rd Qu.:39484
##  Max.   :5104           Max.   :314.53     Max.   :40626
##                                            NA's   :1
##  fossil_fuel_consumption fossil_share_elec fossil_share_energy
##  Min.   :1410            Min.   :35.07     Min.   :73.54
##  1st Qu.:1755            1st Qu.:46.55     1st Qu.:78.76
##  Median :2046            Median :67.38     Median :87.46
##  Mean   :2007            Mean   :61.52     Mean   :84.49
##  3rd Qu.:2370            3rd Qu.:75.02     3rd Qu.:90.02
##  Max.   :2436            Max.   :80.86     Max.   :91.77
##  NA's   :1                                 NA's   :1
##  gas_cons_change_pct gas_cons_change_twh gas_consumption  gas_elec_per_capita
##  Min.   :-16.828     Min.   :-165.743    Min.   : 618.6   Min.   :1248
##  1st Qu.: -5.663     1st Qu.: -50.198    1st Qu.: 763.3   1st Qu.:1590
##  Median : -1.236     Median :  -9.749    Median : 799.0   Median :2117
##  Mean   : -1.643     Mean   : -14.412    Mean   : 847.4   Mean   :2090
##  3rd Qu.:  2.183     3rd Qu.:  19.899    3rd Qu.: 984.9   3rd Qu.:2516
##  Max.   : 10.900     Max.   :  78.529    Max.   :1019.9   Max.   :2841
##  NA's   :1           NA's   :1           NA's   :1
##  gas_electricity gas_energy_per_capita gas_prod_change_pct gas_prod_change_twh
##  Min.   : 86.3   Min.   : 8947         Min.   :-20.411     Min.   :-118.184
##  1st Qu.:104.2   1st Qu.:11372         1st Qu.: -9.519     1st Qu.: -68.901
##  Median :138.8   Median :12125         Median : -3.606     Median : -26.454
##  Mean   :132.8   Mean   :13384         Mean   : -4.388     Mean   : -29.207
##  3rd Qu.:151.4   3rd Qu.:15772         3rd Qu.:  0.351     3rd Qu.:   1.465
##  Max.   :176.2   Max.   :17160         Max.   : 16.262     Max.   :  97.224
##                  NA's   :1             NA's   :1           NA's   :1
##  gas_prod_per_capita gas_production   gas_share_elec  gas_share_energy
##  Min.   : 4443       Min.   : 307.2   Min.   :26.75   Min.   :31.22
##  1st Qu.: 5855       1st Qu.: 391.9   1st Qu.:34.93   1st Qu.:35.01
```

```
##  Median : 6334       Median : 418.8   Median :38.91   Median :35.82
##  Mean   : 9773       Mean   : 609.8   Mean   :37.33   Mean   :35.84
##  3rd Qu.:13718       3rd Qu.: 837.5   3rd Qu.:40.30   3rd Qu.:37.76
##  Max.   :19212       Max.   :1134.6   Max.   :45.97   Max.   :39.90
##  NA's   :1           NA's   :1                        NA's   :1
##  greenhouse_gas_emissions hydro_cons_change_pct hydro_cons_change_twh
##  Min.   : 61.52           Min.   :-32.580       Min.   :-4.3690
##  1st Qu.: 95.34           1st Qu.: -7.455       1st Qu.:-1.1640
##  Median :183.85           Median :  1.619       Median : 0.1050
##  Mean   :156.31           Mean   :  2.307       Mean   :-0.0304
##  3rd Qu.:203.54           3rd Qu.:  9.519       3rd Qu.: 1.2110
##  Max.   :222.21           Max.   : 58.484       Max.   : 5.3930
##  NA's   :1                NA's   :1             NA's   :1
##  hydro_consumption hydro_elec_per_capita hydro_electricity
##  Min.   : 8.795    Min.   : 53.99        Min.   :3.230
##  1st Qu.:13.129    1st Qu.: 80.18        1st Qu.:4.860
##  Median :13.541    Median : 82.02        Median :5.340
##  Mean   :13.425    Mean   : 81.29        Mean   :5.226
##  3rd Qu.:14.127    3rd Qu.: 85.56        3rd Qu.:5.683
##  Max.   :16.983    Max.   :102.15        Max.   :6.880
##  NA's   :1
##  hydro_energy_per_capita hydro_share_elec hydro_share_energy
##  Min.   :147.0           Min.   :0.811    Min.   :0.3300
##  1st Qu.:201.2           1st Qu.:1.248    1st Qu.:0.5000
##  Median :218.2           Median :1.502    Median :0.5780
##  Mean   :209.9           Mean   :1.508    Mean   :0.5813
##  3rd Qu.:220.3           3rd Qu.:1.756    3rd Qu.:0.6750
##  Max.   :252.2           Max.   :2.217    Max.   :0.8650
##  NA's   :1                                NA's   :1
##  low_carbon_cons_change_pct low_carbon_cons_change_twh low_carbon_consumption
##  Min.   :-9.328             Min.   :-39.762            Min.   :210.2
##  1st Qu.:-3.589             1st Qu.:-15.069            1st Qu.:265.2
##  Median : 3.420             Median : 11.363            Median :296.6
##  Mean   : 3.202             Mean   :  8.599            Mean   :350.1
##  3rd Qu.: 7.074             3rd Qu.: 26.773            3rd Qu.:458.5
##  Max.   :20.632             Max.   : 64.262            Max.   :507.1
##  NA's   :1                  NA's   :1                  NA's   :1
##  low_carbon_elec_per_capita low_carbon_electricity low_carbon_energy_per_capita
##  Min.   :1200               Min.   : 74.43         Min.   :3389
##  1st Qu.:1613               1st Qu.: 95.86         1st Qu.:4376
##  Median :1834               Median :117.73         Median :4670
##  Mean   :2016               Mean   :130.78         Mean   :5420
##  3rd Qu.:2562               3rd Qu.:173.71         3rd Qu.:6839
##  Max.   :2730               Max.   :188.77         Max.   :7399
##                                                    NA's   :1
##  low_carbon_share_elec low_carbon_share_energy net_elec_imports
##  Min.   :19.14         Min.   : 8.234          Min.   :-4.300
##  1st Qu.:24.98         1st Qu.: 9.978          1st Qu.: 7.497
##  Median :32.62         Median :12.538          Median :13.015
##  Mean   :38.48         Mean   :15.509          Mean   :13.515
##  3rd Qu.:53.45         3rd Qu.:21.242          3rd Qu.:20.168
##  Max.   :64.93         Max.   :26.457          Max.   :33.190
##                        NA's   :1
##  net_elec_imports_share_demand nuclear_cons_change_pct nuclear_cons_change_twh
```

```
##  Min.   :-1.343                Min.   :-16.727         Min.   :-34.111
##  1st Qu.: 1.860                1st Qu.:-10.070         1st Qu.:-18.372
##  Median : 3.389                Median : -2.491         Median : -8.194
##  Mean   : 3.832                Mean   : -2.826         Mean   : -6.611
##  3rd Qu.: 5.680                3rd Qu.:  2.024         3rd Qu.:  2.604
##  Max.   :10.458                Max.   : 31.650         Max.   : 42.874
##                                NA's   :1               NA's   :1
##  nuclear_consumption nuclear_elec_per_capita nuclear_electricity
##  Min.   : 99.0       Min.   : 523.1          Min.   :36.38
##  1st Qu.:139.1       1st Qu.: 839.4          1st Qu.:53.41
##  Median :178.1       Median :1067.9          Median :69.04
##  Mean   :173.8       Mean   :1035.3          Mean   :65.55
##  3rd Qu.:201.7       3rd Qu.:1203.3          3rd Qu.:74.52
##  Max.   :248.9       Max.   :1519.5          Max.   :90.09
##  NA's   :1
##  nuclear_energy_per_capita nuclear_share_elec nuclear_share_energy
##  Min.   :1432              Min.   :12.43      Min.   :5.164
##  1st Qu.:2233              1st Qu.:15.96      1st Qu.:6.396
##  Median :2724              Median :18.92      Median :7.550
##  Mean   :2756              Mean   :18.35      Mean   :7.280
##  3rd Qu.:3304              3rd Qu.:20.69      3rd Qu.:8.074
##  Max.   :4198              Max.   :23.41      Max.   :9.267
##  NA's   :1                                    NA's   :1
##  oil_cons_change_pct oil_cons_change_twh oil_consumption  oil_elec_per_capita
##  Min.   :-21.2030    Min.   :-179.476    Min.   : 667.0   Min.   :129.8
##  1st Qu.: -2.0770    1st Qu.: -19.831    1st Qu.: 840.3   1st Qu.:145.2
##  Median : -0.7710    Median :  -7.306    Median : 873.9   Median :166.1
##  Mean   : -0.8945    Mean   :  -8.723    Mean   : 869.8   Mean   :173.3
##  3rd Qu.:  1.3120    3rd Qu.:  11.009    3rd Qu.: 947.5   3rd Qu.:198.1
##  Max.   :  7.2240    Max.   :  49.199    Max.   :1001.8   Max.   :238.1
##  NA's   :1           NA's   :1           NA's   :1
##  oil_electricity  oil_energy_per_capita oil_prod_change_pct oil_prod_change_twh
##  Min.   : 8.360   Min.   : 9903         Min.   :-17.454     Min.   :-127.804
##  1st Qu.: 9.688   1st Qu.:12934         1st Qu.: -9.612     1st Qu.: -94.706
##  Median :11.000   Median :13322         Median : -7.614     Median : -45.525
##  Mean   :11.048   Mean   :13704         Mean   : -5.578     Mean   : -49.637
##  3rd Qu.:12.220   3rd Qu.:15792         3rd Qu.: -1.716     3rd Qu.:  -8.538
##  Max.   :14.060   Max.   :16532         Max.   : 13.424     Max.   :  62.335
##                   NA's   :1             NA's   :1           NA's   :1
##  oil_prod_per_capita oil_production   oil_share_elec  oil_share_energy
##  Min.   : 5113       Min.   : 353.5   Min.   :2.333   Min.   :33.98
##  1st Qu.: 8056       1st Qu.: 518.2   1st Qu.:2.795   1st Qu.:35.83
##  Median : 8980       Median : 602.6   Median :3.092   Median :37.02
##  Mean   :11992       Mean   : 749.8   Mean   :3.141   Mean   :36.95
##  3rd Qu.:14588       3rd Qu.: 890.6   3rd Qu.:3.436   3rd Qu.:37.53
##  Max.   :24861       Max.   :1468.2   Max.   :4.362   Max.   :39.56
##  NA's   :1           NA's   :1                        NA's   :1
##  other_renewable_consumption other_renewable_electricity
##  Min.   : 12.13              Min.   : 3.88
##  1st Qu.: 28.99              1st Qu.: 9.29
##  Median : 46.06              Median :16.42
##  Mean   : 63.58              Mean   :21.15
##  3rd Qu.:106.56              3rd Qu.:34.87
##  Max.   :125.30              Max.   :41.21
```

```
##  NA's   :1
##  other_renewable_exc_biofuel_electricity other_renewables_cons_change_pct
##  Min.   :0.000                           Min.   :-10.385
##  1st Qu.:0.000                           1st Qu.:  3.483
##  Median :0.000                           Median : 10.697
##  Mean   :0.001                           Mean   : 10.755
##  3rd Qu.:0.000                           3rd Qu.: 17.586
##  Max.   :0.010                           Max.   : 29.343
##  NA's   :6                               NA's   :1
##  other_renewables_cons_change_twh other_renewables_elec_per_capita
##  Min.   :-12.983                  Min.   : 65.7
##  1st Qu.:  1.732                  1st Qu.:151.6
##  Median :  3.417                  Median :255.8
##  Mean   :  4.583                  Mean   :320.3
##  3rd Qu.:  5.727                  3rd Qu.:518.5
##  Max.   : 20.743                  Max.   :592.5
##  NA's   :1
##  other_renewables_elec_per_capita_exc_biofuel
##  Min.   :0.00000
##  1st Qu.:0.00000
##  Median :0.00000
##  Mean   :0.01495
##  3rd Qu.:0.00000
##  Max.   :0.15000
##  NA's   :6
##  other_renewables_energy_per_capita other_renewables_share_elec
##  Min.   : 205.4                     Min.   : 1.029
##  1st Qu.: 473.5                     1st Qu.: 2.339
##  Median : 720.1                     Median : 4.551
##  Mean   : 966.8                     Mean   : 6.427
##  3rd Qu.:1551.4                     3rd Qu.:10.915
##  Max.   :1847.5                     Max.   :14.108
##  NA's   :1
##  other_renewables_share_elec_exc_biofuel other_renewables_share_energy
##  Min.   :0e+00                           Min.   :0.456
##  1st Qu.:0e+00                           1st Qu.:1.085
##  Median :0e+00                           Median :1.947
##  Mean   :3e-04                           Mean   :2.923
##  3rd Qu.:0e+00                           3rd Qu.:4.923
##  Max.   :3e-03                           Max.   :6.537
##  NA's   :6                               NA's   :1
##  per_capita_electricity primary_energy_consumption renewables_cons_change_pct
##  Min.   :4110           Min.   :1930               Min.   :-17.055
##  1st Qu.:4907           1st Qu.:2246               1st Qu.:  3.108
##  Median :5627           Median :2375               Median : 13.704
##  Mean   :5559           Mean   :2369               Mean   : 13.212
##  3rd Qu.:6432           3rd Qu.:2631               3rd Qu.: 19.974
##  Max.   :6657           Max.   :2713               Max.   : 54.688
##                         NA's   :1                  NA's   :1
##  renewables_cons_change_twh renewables_consumption renewables_elec_per_capita
##  Min.   :-29.271            Min.   : 28.02         Min.   : 160.9
##  1st Qu.:  1.725            1st Qu.: 54.69         1st Qu.: 302.3
##  Median :  9.783            Median :115.19         Median : 735.6
##  Mean   : 15.210            Mean   :176.28         Mean   : 980.8
```

```
##  3rd Qu.: 32.627           3rd Qu.:311.60        3rd Qu.:1750.7
##  Max.   : 48.382           Max.   :408.15        Max.   :2191.0
##  NA's   :1                 NA's   :1
##  renewables_electricity renewables_energy_per_capita renewables_share_elec
##  Min.   :  9.54         Min.   : 472.5               Min.   : 2.479
##  1st Qu.: 18.50         1st Qu.: 895.8               1st Qu.: 4.657
##  Median : 47.23         Median :1801.0               Median :13.093
##  Mean   : 65.23         Mean   :2663.8               Mean   :20.134
##  3rd Qu.:117.34         3rd Qu.:4667.9               3rd Qu.:35.697
##  Max.   :152.39         Max.   :5903.4               Max.   :52.049
##                         NA's   :1
##  renewables_share_energy solar_cons_change_pct solar_cons_change_twh
##  Min.   : 1.043          Min.   : -3.335       Min.   :-1.107
##  1st Qu.: 2.047          1st Qu.:  6.519       1st Qu.: 0.003
##  Median : 4.869          Median : 27.273       Median : 0.052
##  Mean   : 8.228          Mean   : 70.409       Mean   : 1.443
##  3rd Qu.:13.980          3rd Qu.: 85.810       3rd Qu.: 2.493
##  Max.   :21.293          Max.   :504.968       Max.   : 8.755
##  NA's   :1               NA's   :1             NA's   :1
##  solar_consumption solar_elec_per_capita solar_electricity
##  Min.   : 0.003    Min.   :  0.0000      Min.   : 0.000
##  1st Qu.: 0.029    1st Qu.:  0.1643      1st Qu.: 0.010
##  Median : 3.489    Median : 26.1615      Median : 1.680
##  Mean   :12.783    Mean   : 84.4646      Mean   : 5.702
##  3rd Qu.:29.873    3rd Qu.:183.6140      3rd Qu.:12.348
##  Max.   :36.072    Max.   :277.7800      Max.   :19.320
##  NA's   :1
##  solar_energy_per_capita solar_share_elec solar_share_energy
##  Min.   :  0.047         Min.   :0.000    Min.   :0.0000
##  1st Qu.:  0.482         1st Qu.:0.003    1st Qu.:0.0010
##  Median : 54.551         Median :0.466    Median :0.1470
##  Mean   :190.235         Mean   :1.813    Mean   :0.6152
##  3rd Qu.:441.461         3rd Qu.:3.796    3rd Qu.:1.4140
##  Max.   :521.737         Max.   :6.599    Max.   :1.8820
##  NA's   :1                                NA's   :1
##  wind_cons_change_pct wind_cons_change_twh wind_consumption
##  Min.   :-14.15       Min.   :-26.813      Min.   :  2.628
##  1st Qu.: 10.82       1st Qu.:  1.745      1st Qu.: 11.326
##  Median : 24.06       Median :  4.266      Median : 51.152
##  Mean   : 21.58       Mean   :  8.106      Mean   : 77.452
##  3rd Qu.: 33.59       3rd Qu.: 16.446      3rd Qu.:141.561
##  Max.   : 55.20       Max.   : 37.982      Max.   :205.019
##  NA's   :1            NA's   :1            NA's   :1
##  wind_elec_per_capita wind_electricity wind_energy_per_capita wind_share_elec
##  Min.   :  15.92      Min.   : 0.940   Min.   :  44.49        Min.   : 0.249
##  1st Qu.:  73.25      1st Qu.: 4.482   1st Qu.: 185.52        1st Qu.: 1.129
##  Median : 375.63      Median :24.120   Median : 799.75        Median : 6.689
##  Mean   : 494.75      Mean   :33.152   Mean   :1161.34        Mean   :10.386
##  3rd Qu.: 926.58      3rd Qu.:62.108   3rd Qu.:2120.69        3rd Qu.:18.897
##  Max.   :1240.95      Max.   :86.310   Max.   :2965.35        Max.   :29.479
##                                        NA's   :1
##  wind_share_energy
##  Min.   : 0.099
##  1st Qu.: 0.424
```

```
##  Median : 2.162
##  Mean   : 3.679
##  3rd Qu.: 6.351
##  Max.   :10.696
##  NA's   :1
```

```
# Total number of missing values
sum(is.na(uk_energy))
```

```
## [1] 105
```

The above code chunk has revealed the type of variables that make up the uk_energy dataset, the missing values and the summary statistics of each numerical variable. It was also observed that the missing values have now reduced to 105.

From the research questions, it is clear that we only need the following variables to answer our research questions above.

- year
- gdp
- coal_share_elec
- gas_share_elec
- oil_share_elec
- fossil_share_elec
- nuclear_share_elec
- wind_share_elec
- solar_share_elec
- hydro_share_elec
- biofuel_share_elec
- carbon_intensity_elec
- energy_per_capita

Therefore, we only select the desired variables in the code chunk below

```
uk_electricity <- uk_energy %>%
  select(
      year,
      gdp,

      # Fossil fuels (electricity-based)
      coal_share_elec,
      gas_share_elec,
      oil_share_elec,

      # Overall mix
      fossil_share_elec,

      # Low-carbon breakdown
      nuclear_share_elec,
      wind_share_elec,
      solar_share_elec,
      hydro_share_elec,
      biofuel_share_elec,
```

```r
    # Outcomes
    carbon_intensity_elec,
    energy_per_capita
  )

glimpse(uk_electricity)
```

```
## Rows: 26
## Columns: 13
## $ year                  <dbl> 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, ~
## $ gdp                   <dbl> 1.889565e+12, 1.939268e+12, 1.983994e+12, 2.0506~
## $ coal_share_elec       <dbl> 31.812, 34.166, 32.094, 34.771, 33.456, 33.799, ~
## $ gas_share_elec        <dbl> 39.272, 36.882, 39.324, 37.387, 39.871, 38.318, ~
## $ oil_share_elec        <dbl> 3.729, 3.059, 3.024, 2.885, 2.775, 3.143, 3.534,~
## $ fossil_share_elec     <dbl> 74.813, 74.107, 74.442, 75.043, 76.102, 75.260, ~
## $ nuclear_share_elec    <dbl> 22.559, 23.414, 22.686, 22.272, 20.309, 20.490, ~
## $ wind_share_elec       <dbl> 0.249, 0.249, 0.323, 0.324, 0.492, 0.728, 1.062,~
## $ solar_share_elec      <dbl> 0.000, 0.000, 0.000, 0.000, 0.000, 0.003, 0.003,~
## $ hydro_share_elec      <dbl> 1.350, 1.053, 1.237, 0.811, 1.229, 1.235, 1.155,~
## $ biofuel_share_elec    <dbl> 1.029, 1.177, 1.312, 1.549, 1.868, 2.284, 2.336,~
## $ carbon_intensity_elec <dbl> 521.99, 528.99, 520.89, 536.78, 536.02, 535.26, ~
## $ energy_per_capita     <dbl> 45110.68, 45402.14, 44185.53, 44612.46, 44717.52~
```

From the above code chunk, it was observed that the number of columns and variables in our desired data set is 13 columns and 26 rows. We have selected variables that showcase the fossil fuels share of electricity generated, Low-carbon breakdown and outcomes of the electricity generated.

To get a clearer picture of the type of variables that make up the data set, there is need to apply "sapply" function which will print the data class of the respective variable

```r
# Print the class of each column in uk_electricity
sapply(uk_electricity, class)
```

```
##                  year                   gdp       coal_share_elec
##             "numeric"             "numeric"             "numeric"
##        gas_share_elec        oil_share_elec     fossil_share_elec
##             "numeric"             "numeric"             "numeric"
##    nuclear_share_elec       wind_share_elec      solar_share_elec
##             "numeric"             "numeric"             "numeric"
##      hydro_share_elec    biofuel_share_elec carbon_intensity_elec
##             "numeric"             "numeric"             "numeric"
##     energy_per_capita
##             "numeric"
```

4.3 Characteristics of Variables of Interest

- **fossil__share__elec** (share variable, %): This variable captures the overall contribution of fossil fuels (coal, gas, and oil) to electricity generation and serves as the primary indicator for RQ1. It provides a high-level view of the decline of fossil fuels in the power sector and enables direct comparison with low-carbon sources over time.

- **coal_share_elec, gas_share_elec, and oil_share_elec** (share variables, %): These variables disaggregate fossil fuel electricity into its key components, allowing for a more nuanced analysis of the transition. Coal is particularly important due to its high carbon intensity, while natural gas often acts as a transitional fuel. Although oil contributes only a small share of electricity generation, it is retained to demonstrate its negligible and declining role, ensuring completeness in the fossil fuel category.
- **nuclear_share_elec, wind_share_elec, solar_share_elec, hydro_share_elec, and biofuel_share_elec** (share variables, %): These variables collectively represent the low-carbon electricity mix and are central to RQ2. Low-carbon energy is defined here as the combination of nuclear and renewable sources. Disaggregating these components allows for deeper insight into which technologies are driving the transition, revealing, for example, the rapid growth of wind and solar relative to more stable sources such as nuclear and hydro.

It is important to note that while biofuels are classified as renewable, they differ from other renewables in that they involve combustion and therefore produce direct emissions, highlighting that not all low-carbon sources are emission-free.

- **carbon_intensity_elec** (intensity variable, gCO /kWh): This variable provides a direct measure of the environmental performance of the electricity system and is a key indicator of decarbonisation. Unlike share-based variables, it captures the combined effect of changes in the energy mix and improvements in generation efficiency, making it highly relevant for evaluating the overall impact of the transition (Bruns et al., 2020).
- **energy_per_capita** (per-capita consumption variable, kWh): This variable addresses RQ3 by capturing trends in overall energy demand while controlling for population size. It enables the analysis to distinguish between structural changes in energy use and fluctuations driven by external factors such as economic cycles or shocks (e.g., the COVID-19 pandemic). However, as it reflects total energy consumption rather than electricity alone, its interpretation should be made cautiously in the context of an electricity-focused analysis.
- **gdp** (economic indicator, constant USD): Gross Domestic Product is included to provide economic context for changes in energy demand and electricity consumption. It supports the interpretation of RQ3 by helping to distinguish whether observed changes in energy use are associated with economic growth, structural shifts in the economy, or efficiency improvements.

5 Exploration of Various variables in UK Electricity Data Set

The dataset used in this analysis is a time series covering the period from 2000 to 2025, with 13 variables capturing changes in electricity generation, economic activity, and environmental outcomes. Before directly addressing the research questions, it is important to first explore how these variables behave over time. This helps to identify key trends, spot any irregularities, and guide the choice of appropriate visualisations.

Since the data is time-based, the analysis focuses on how values evolve over time, including long-term trends, turning points, and shifts in relationships between variables. As a result, the exploratory stage relies mainly on trend-based and comparative visualisations, which provide a clear view of how the UK's electricity system has changed.

Each visualisation is designed with a specific purpose in mind, linking observed patterns in the data to broader structural changes in the UK electricity sector and directly supporting the research questions.

5.1 Fossil Fuels vs. Low-Carbon Sources

To examine overall electricity trends in the UK, this section compares how fossil fuels and low-carbon sources have evolved over time. This begins by grouping the variables that make up low-carbon sources into a single combined measure.

The code chunk below combines the nuclear,wind,solar,hydro and biofuel to form low_carbon_share

```
# We clean our data by grouping together variables that fall under Low Carbon Share.
uk_electricity <- uk_electricity %>%
  mutate(
    low_carbon_share = nuclear_share_elec +
                       wind_share_elec +
                       solar_share_elec +
                       hydro_share_elec +
                       biofuel_share_elec
  )
```

```
summary_tbl <- uk_electricity %>%
  select(
    `Fossil share of electricity (%)` = fossil_share_elec,
    `Coal share of electricity (%)`   = coal_share_elec,
    `Low-carbon share (%)`            = low_carbon_share,
    `Carbon intensity (gCO2/kWh)`     = carbon_intensity_elec,
    `Energy per capita (kWh)`         = energy_per_capita
  ) %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "value") %>%
  group_by(Variable) %>%
  summarise(
    Min    = round(min(value, na.rm = TRUE), 1),
    Mean   = round(mean(value, na.rm = TRUE), 1),
    Median = round(median(value, na.rm = TRUE), 1),
    Max    = round(max(value, na.rm = TRUE), 1),
    SD     = round(sd(value, na.rm = TRUE), 1)
  )

kable(summary_tbl,
      caption = "Table 2. Descriptive statistics for key variables (2000-2025).")
```

Table 2: Table 2. Descriptive statistics for key variables (2000–2025).

Variable	Min	Mean	Median	Max	SD
Carbon intensity (gCO2/kWh)	216.5	419.2	502.1	559.3	132.1
Coal share of electricity (%)	0.1	21.0	28.8	39.2	14.9
Energy per capita (kWh)	28015.8	37347.2	37094.2	45402.1	5962.6
Fossil share of electricity (%)	35.1	61.5	67.4	80.9	15.6
Low-carbon share (%)	19.1	38.5	32.6	64.9	15.6

Key observations from Table 2:

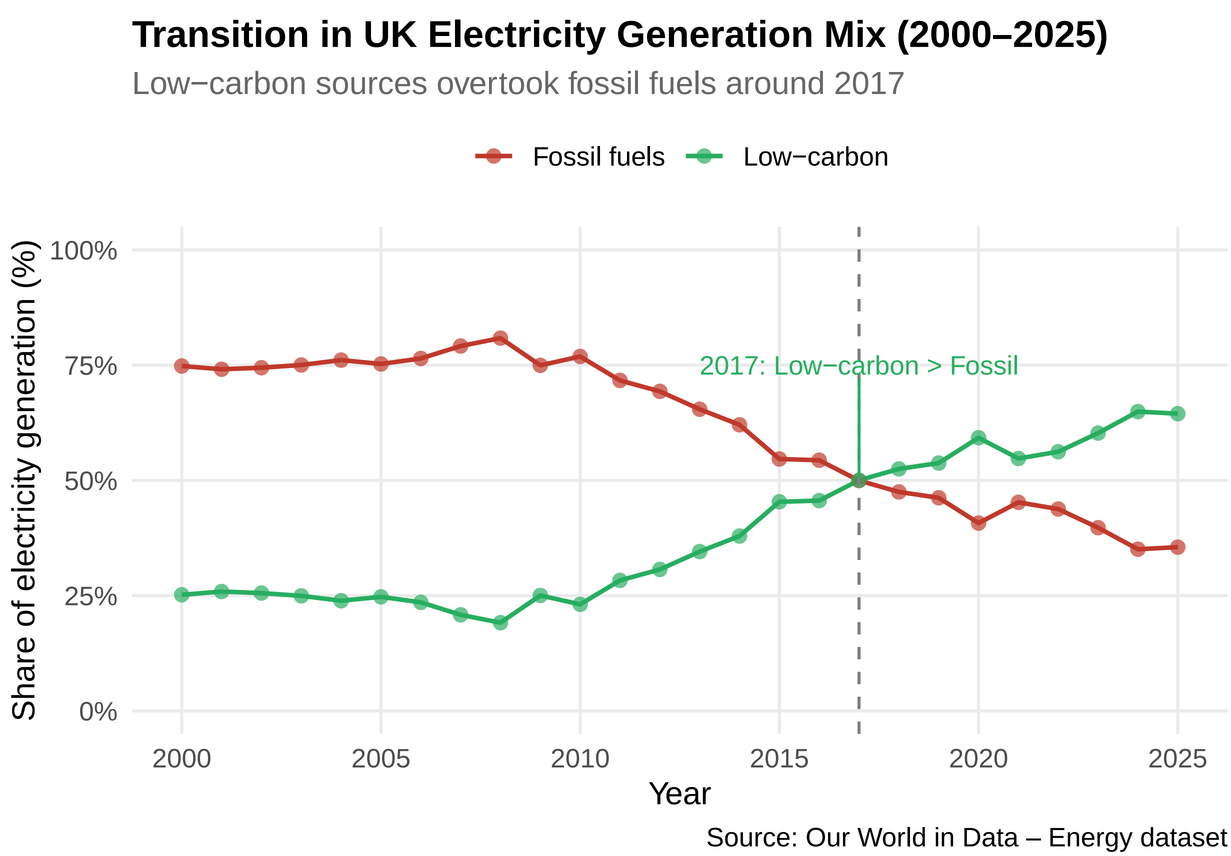
- Fossil share of electricity ranges from approximately 35% to 81%, a 46-percentage-point range with a high standard deviation, consistent with sustained structural change rather than cyclical variation. Coal share ranges from under 1% to over 39%, confirming near-complete elimination by the end of the study period.
- Low-carbon share ranges from approximately 19% to 65%, with the upward trajectory being the mirror image of the fossil decline.

- Carbon intensity has a mean of approximately 404 gCO /kWh but a 2025 value of 217 gCO /kWh, confirming that the most dramatic decarbonisation occurred in the latter half of the period.
- Energy per capita shows a large standard deviation relative to its range, indicating substantial year-to-year variation that warrants caution when attributing the decline to any single cause.

To understand the overall direction of change in the UK electricity system, a line plot is used to visualise these trends over time.

```r
label_years <- seq(2000, 2025, 5)

uk_trend <- uk_electricity %>%
  mutate(label = if_else(year %in% label_years,
                         as.character(year), NA_character_))

# Extract 2017 point for annotation
label_2017 <- uk_trend %>%
  filter(year == 2017)

ggplot(uk_trend, aes(x = year)) +

  geom_line(aes(y = fossil_share_elec, colour = "Fossil fuels"),
            linewidth = 0.8) +

  geom_line(aes(y = low_carbon_share, colour = "Low-carbon"),
            linewidth = 0.8) +

  geom_point(aes(y = fossil_share_elec, colour = "Fossil fuels"),
           size = 2, alpha = 0.7) +

geom_point(aes(y = low_carbon_share, colour = "Low-carbon"),
           size = 2, alpha = 0.7) +

  # Highlight crossover with vertical line
  geom_vline(xintercept = 2017, linetype = "dashed", colour = "grey50") +

  # Label directly on the line (2017)
  geom_text_repel(
    data = label_2017,
    aes(y = low_carbon_share,
        label = "2017: Low-carbon > Fossil"),
    nudge_y = 25,
    colour = "#27ae60",
    size = 3.5,

  ) +

  scale_colour_manual(values = c(
    "Fossil fuels" = "#c0392b",
    "Low-carbon" = "#27ae60"
  )) +

  scale_x_continuous(breaks = seq(2000, 2025, 5)) +
  scale_y_continuous(labels = scales::label_percent(scale = 1),
                     limits = c(0, 100)) +
```

```r
labs(
  title    = "Transition in UK Electricity Generation Mix (2000-2025)",
  subtitle = "Low-carbon sources overtook fossil fuels around 2017",
  x        = "Year",
  y        = "Share of electricity generation (%)",
  colour   = NULL,
  caption  = "Source: Our World in Data - Energy dataset"
) +

theme_minimal(base_size = 12) +
theme(
  plot.title       = element_text(face = "bold", size = 14),
  plot.subtitle    = element_text(colour = "grey40"),
  panel.grid.minor = element_blank(),
  legend.position  = "top"
)
```



The evolution of the UK electricity sector between 2000 and 2025 shows a clear and gradual shift toward decarbonisation, rather than a single sudden change. In the early period (roughly 2000 to 2008), there is little evidence of progress. Fossil fuels consistently dominate electricity generation, accounting for around 75–80% of the mix, while low-carbon sources remain relatively flat or even decline slightly. This suggests that early climate policies had limited immediate impact, and that meaningful change had not yet begun.

The transition becomes more visible in the 2010s, particularly after 2012, when the pace of change accelerates. During this period, fossil fuel use declines rapidly while low-carbon sources expand significantly, indicating a structural shift in how electricity is generated. This aligns with the combined effects of policy interventions, such as carbon pricing, and the rapid growth of renewable technologies like wind.

A key turning point occurs around 2016–2017, when low-carbon electricity overtakes fossil fuels for the first time. This marks a fundamental shift in the UK's power system, where cleaner sources become the dominant form of generation. While there is some short-term variation around 2020, partly influenced by the COVID-19 pandemic and reduced electricity demand, the overall trend remains consistent, and the system quickly returns to its longer-term trajectory.

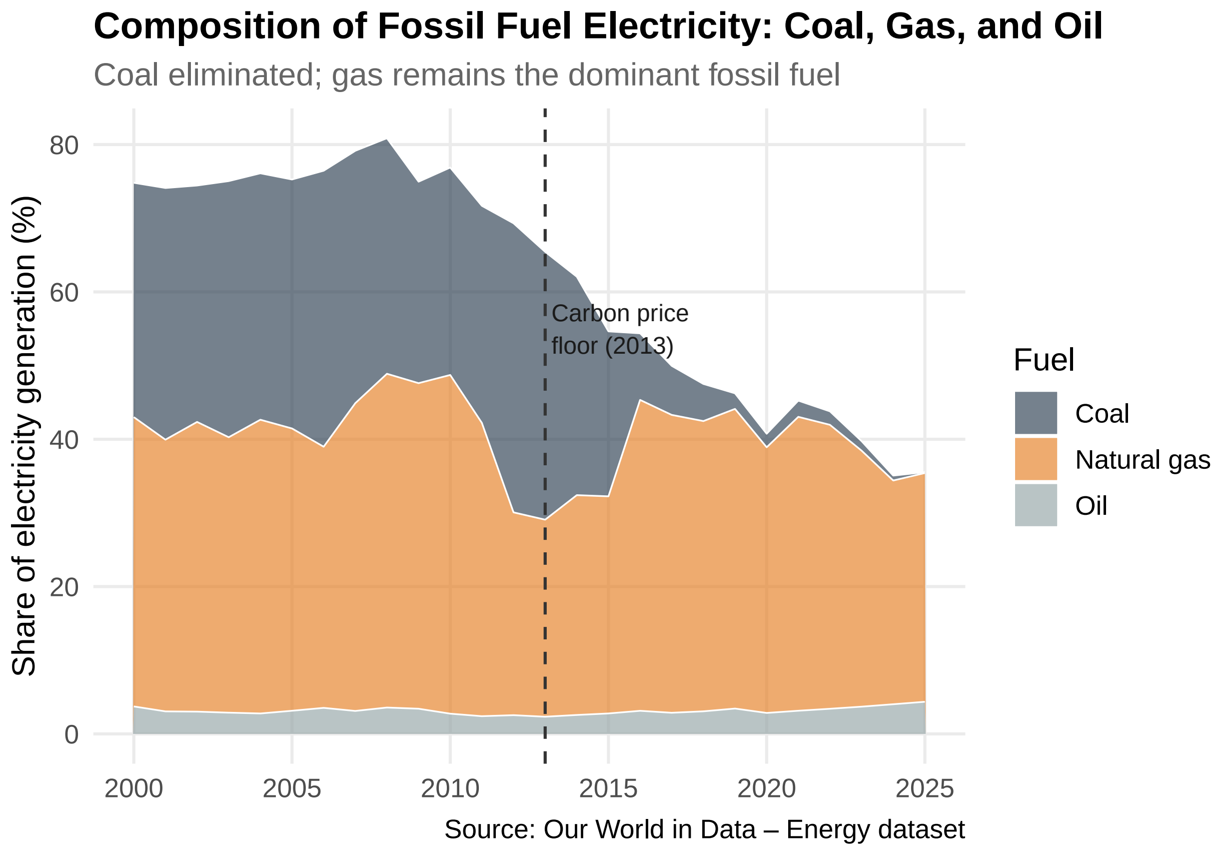
By 2025, the UK appears to have entered a new phase, where low-carbon sources make up the majority of electricity generation, stabilising at around 60–65%. At the same time, fossil fuels take on a more limited, supporting role rather than being the primary source of power. This shift highlights that decarbonisation in the UK electricity sector is not just a temporary trend, but a sustained and structural transformation.

This supports Research Question 1 by showing the overall decline in fossil fuel electricity and the point at which low-carbon sources became dominant in the UK electricity mix.

5.2 Internal Dynamics of Fossil Fuel Decline

While Visualisation 1 shows the overall fossil vs low-carbon split, it does not reveal how coal and gas have behaved differently within the fossil category. A stacked area chart for the three fossil fuels allows direct comparison of their relative contributions over time, making their divergent trajectories immediately apparent.

```r
uk_fossil_long <- uk_electricity %>%
  select(year, coal_share_elec, gas_share_elec, oil_share_elec) %>%
  pivot_longer(-year, names_to = "source", values_to = "share") %>%
  mutate(source = recode(source,
    coal_share_elec = "Coal",
    gas_share_elec  = "Natural gas",
    oil_share_elec  = "Oil"
  ))

ggplot(uk_fossil_long, aes(x = year, y = share, fill = source)) +
  geom_area(alpha = 0.65, colour = "white", linewidth = 0.3) +
  scale_fill_manual(values = c(
    "Coal"        = "#2c3e50",
    "Natural gas" = "#e67e22",
    "Oil"         = "#95a5a6"
  )) +
  scale_x_continuous(breaks = seq(2000, 2025, 5)) +
  geom_vline(xintercept = 2013, linetype = "dashed", colour = "grey20") +
  annotate("text", x = 2013.2, y = 55, label = "Carbon price\nfloor (2013)",
           size = 3, colour = "grey10", hjust = 0) +
  labs(
    title    = "Composition of Fossil Fuel Electricity: Coal, Gas, and Oil",
    subtitle = "Coal eliminated; gas remains the dominant fossil fuel",
    x = NULL, y = "Share of electricity generation (%)", fill = "Fuel",
    caption = "Source: Our World in Data - Energy dataset"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(colour = "grey40"),
        panel.grid.minor = element_blank())
```



The composition of fossil fuel electricity in the UK reveals two very different stories: the rapid collapse of coal and the continued importance of natural gas. In the early 2000s, coal and gas contributed similar shares of electricity generation, with small fluctuations largely driven by fuel prices. However, this balance begins to break down around 2010, where coal enters a clear and sustained decline.

One important factor behind this shift is the change in relative fuel costs. Around 2010, global gas prices increased, making gas-fired generation more expensive and temporarily slowing its growth. This helps explain the visible dip and volatility in gas share during this period. However, despite these short-term fluctuations, the long-term trend shows gas becoming the dominant fossil fuel as coal declines.

The most significant turning point occurs after 2013 with the introduction of the Carbon Price Floor. This policy effectively placed a minimum price on carbon emissions in the UK power sector. In simple terms, it made it more expensive for power plants to emit CO . Since coal produces much more carbon per unit of electricity than gas, it became significantly more costly to run coal plants compared to gas plants.

As a result, electricity generators shifted away from coal very quickly. This is clearly visible in the plot, where coal's share drops sharply after 2013 and never recovers. The policy did not start the decline entirely, coal was already weakening, but it accelerated the transition by making coal economically unviable in most cases.

Meanwhile, natural gas takes on a more stable but complex role. As coal disappears, gas becomes the primary remaining fossil fuel, effectively acting as a "bridge" between a high-carbon system and a low-carbon one. However, its persistence also highlights a limitation of the transition: by 2025, the UK still depends heavily on gas for reliable electricity generation, particularly when renewable output is variable.

Oil, in contrast, plays a negligible role throughout the entire period and has little impact on overall trends.

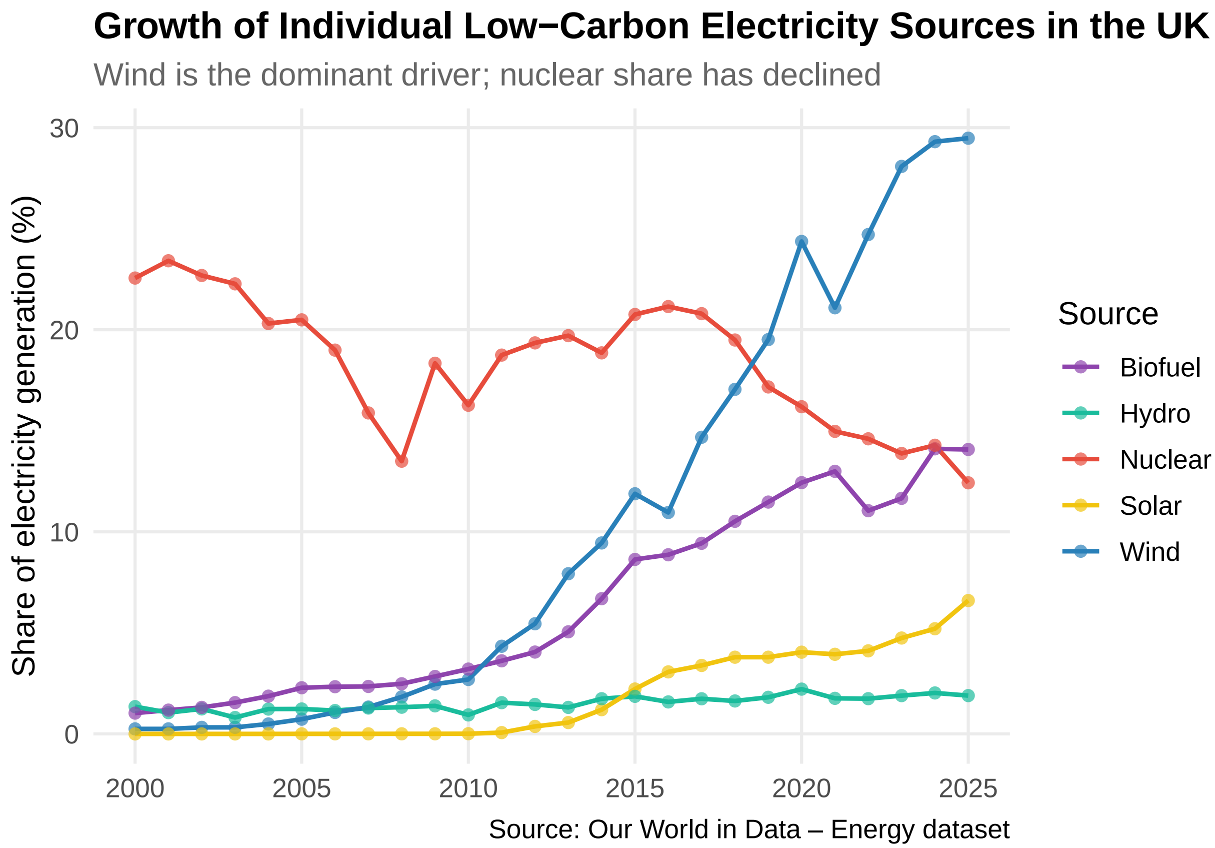
Overall, this analysis shows that the UK's reduction in fossil fuel electricity is largely driven by the elimination of coal rather than a complete phase-out of fossil fuels. While this represents significant progress, the continued reliance on natural gas suggests that the transition remains incomplete and dependent on fossil-based support systems.

This directly addresses Research Question 1 by showing that the decline in fossil fuel electricity is primarily driven by the near-elimination of coal, while natural gas remains structurally embedded in the system.

5.3 Growth of Low-Carbon Energy Sources

The aggregate low-carbon share in Visualisation 1 conceals which technologies are driving growth. A multi-line chart for each low-carbon source reveals their individual trajectories, enabling a more granular answer to RQ2 about which technologies have driven the transition.

```r
uk_lowcarbon_long <- uk_electricity %>%
  select(year, wind_share_elec, solar_share_elec,
         hydro_share_elec, biofuel_share_elec, nuclear_share_elec) %>%
  pivot_longer(-year, names_to = "source", values_to = "share") %>%
  mutate(source = recode(source,
    wind_share_elec    = "Wind",
    solar_share_elec   = "Solar",
    hydro_share_elec   = "Hydro",
    biofuel_share_elec = "Biofuel",
    nuclear_share_elec = "Nuclear"
  ))

ggplot(uk_lowcarbon_long, aes(x = year, y = share,
                              colour = source, group = source)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.6, alpha = 0.7) +
  scale_colour_manual(values = c(
    "Wind"    = "#2980b9",
    "Solar"   = "#f1c40f",
    "Hydro"   = "#1abc9c",
    "Biofuel" = "#8e44ad",
    "Nuclear" = "#e74c3c"
  )) +
  scale_x_continuous(breaks = seq(2000, 2025, 5)) +
  labs(
    title    = "Growth of Individual Low-Carbon Electricity Sources in the UK",
    subtitle = "Wind is the dominant driver; nuclear share has declined",
    x = NULL, y = "Share of electricity generation (%)", colour = "Source",
    caption = "Source: Our World in Data - Energy dataset"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(colour = "grey40"),
        panel.grid.minor = element_blank())
```



The breakdown of low-carbon electricity sources shows that the UK's energy transition is not the result of all technologies growing evenly, but rather a shift driven by a few key technologies, most notably wind power.

The most striking trend is the rapid rise of wind energy. From almost zero in 2000, wind grows steadily and then accelerates sharply after 2010, becoming the dominant source of low-carbon electricity by 2025. This reflects both technological improvements and strong policy support, particularly the expansion of offshore wind. In effect, wind has become the main driver of the UK's decarbonisation, replacing older forms of low-carbon generation.

In contrast, nuclear power, once the backbone of low-carbon electricity, shows a gradual decline over time. While it remains an important contributor, its share decreases due to the aging of existing plants and the lack of large-scale new capacity. This highlights a key structural shift: the UK is moving away from stable, continuous (baseload) low-carbon generation toward more variable renewable sources.

Other technologies provide additional insight into how the transition has unfolded. Biofuels increase rapidly between around 2012 and 2021, likely reflecting the conversion of former coal plants such as Drax, before stabilising at a moderate level. Solar energy, although smaller in overall share, shows fast growth after 2010, driven by falling technology costs and supportive policies such as feed-in tariffs. Meanwhile, hydro remains relatively constant throughout the period, reflecting geographical limits rather than policy or technological change.

Overall, the transition to low-carbon electricity in the UK is largely a wind-led transformation, supported by a mix of other renewable sources. However, this shift also introduces new challenges. Unlike nuclear, which provides consistent output, wind and solar are variable, meaning their generation depends on weather conditions. As a result, the increasing reliance on these sources implies a need for improved energy storage, grid flexibility, and backup capacity to ensure system reliability.

This answers Research Question 2 by demonstrating that the growth in low-carbon electricity is largely driven by wind power, which has overtaken other technologies and enabled low-carbon sources to surpass fossil fuels.

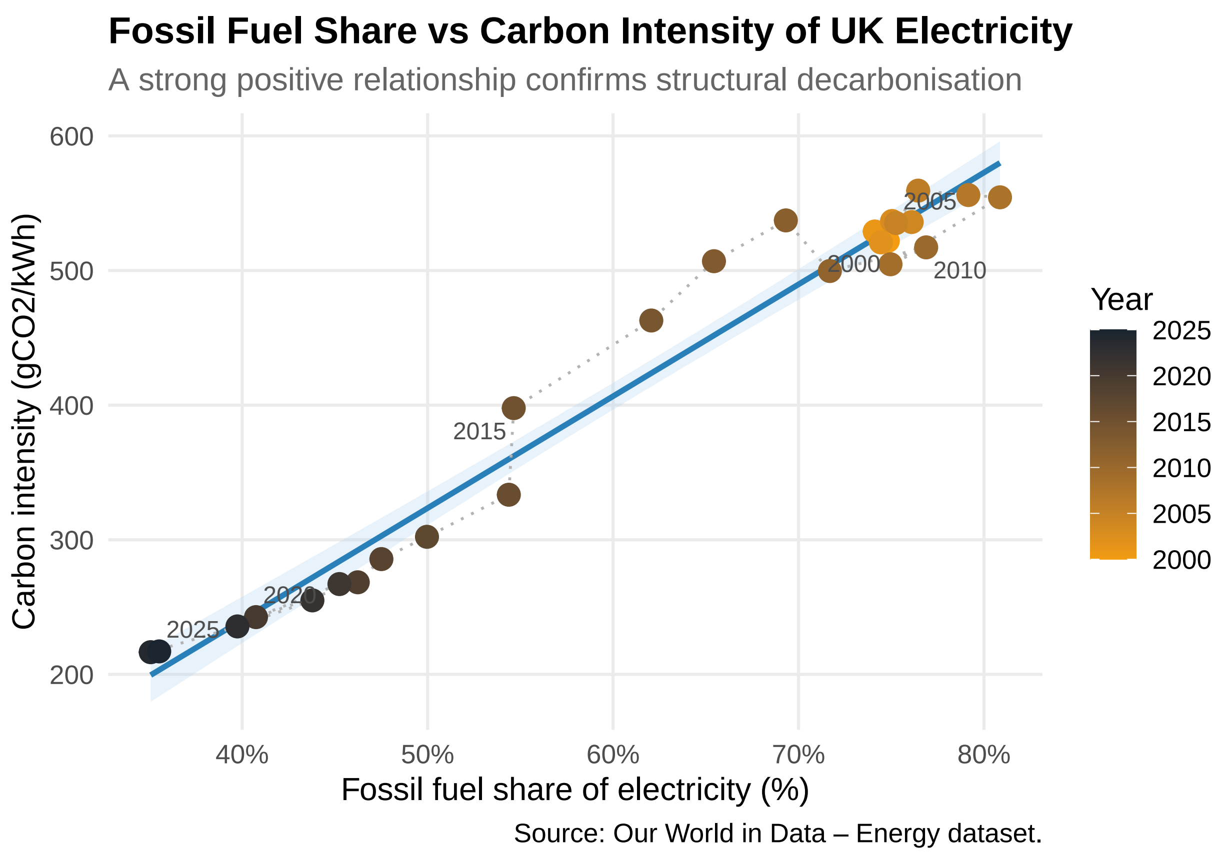
5.4 Environmental Impact and Structural Decoupling

A scatter plot with a regression line is the most appropriate visualisation for exploring the relationship between two continuous variables. Colouring points by year adds a temporal dimension, showing not only whether the two variables are correlated but whether the relationship has been consistent across the full 26-year period or has shifted over time.

```r
label_years <- seq(2000, 2025, 5)

uk_scatter <- uk_electricity %>%
  filter(!is.na(carbon_intensity_elec), !is.na(fossil_share_elec)) %>%
  mutate(label = if_else(year %in% label_years,
                         as.character(year), NA_character_))

ggplot(uk_scatter, aes(x = fossil_share_elec, y = carbon_intensity_elec)) +
  geom_smooth(method = "lm", se = TRUE, colour = "#2980b9",
              fill = "#aed6f1", alpha = 0.3, linewidth = 1) +
  geom_path(colour = "grey70", linewidth = 0.5, linetype = "dotted") +
  geom_point(aes(colour = year), size = 3.5) +
  geom_text_repel(aes(label = label), size = 3, colour = "grey30",
                  box.padding = 0.3, max.overlaps = 10) +
  scale_colour_gradient(low = "#f39c12", high = "#1a252f", name = "Year") +
  scale_x_continuous(labels = label_percent(scale = 1)) +
  labs(
    title    = "Fossil Fuel Share vs Carbon Intensity of UK Electricity",
    subtitle = "A strong positive relationship confirms structural decarbonisation",
    x        = "Fossil fuel share of electricity (%)",
    y        = "Carbon intensity (gCO /kWh)",
    caption  = "Source: Our World in Data - Energy dataset."
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        plot.subtitle = element_text(colour = "grey40"),
        panel.grid.minor = element_blank(),
        legend.position = "right")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 20 rows containing missing values or values outside the scale range
## (`geom_text_repel()`).
```



The strong positive relationship between fossil fuel share and carbon intensity provides clear evidence of the UK's progress toward decarbonisation. As the share of electricity generated from fossil fuels declines, carbon intensity falls in a consistent and predictable way. In practical terms, this means that reducing reliance on fossil fuels has directly translated into lower emissions across the electricity system.

The plot also shows how this transition has unfolded over time. Earlier years (around 2000) are clustered at the top-right of the chart, reflecting both high fossil fuel dependence and high carbon intensity (around 500 gCO /kWh). Over time, the data points move steadily toward the bottom-left, reaching much lower levels of carbon intensity by 2025 (close to 200 gCO /kWh). The dotted path connecting the points highlights that this shift has been gradual and consistent rather than erratic, suggesting a sustained structural change rather than short-term fluctuations.

A closer look at the early period (2000–2010) shows relatively little movement. During this time, fossil fuel dependence remained high and reductions in carbon intensity were limited. This indicates that meaningful progress had not yet begun. In contrast, the period after 2010 shows much larger year-to-year changes, with data points spreading out more clearly along the downward trend. This suggests that the decarbonisation process accelerated significantly in the 2010s, likely driven by the phase-out of coal and the rapid expansion of renewable energy, particularly wind.

Overall, the stability of the relationship between fossil fuel share and carbon intensity indicates that the UK's progress is not accidental or cyclical, but structural. The continued alignment of recent data points with the overall trend also suggests that reducing fossil fuel use remains an effective and reliable way to lower emissions in the power sector.

This directly addresses Resaerch Question 3 by confirming that the decline in carbon intensity reflects a structural change in the electricity system, rather than short-term fluctuations.

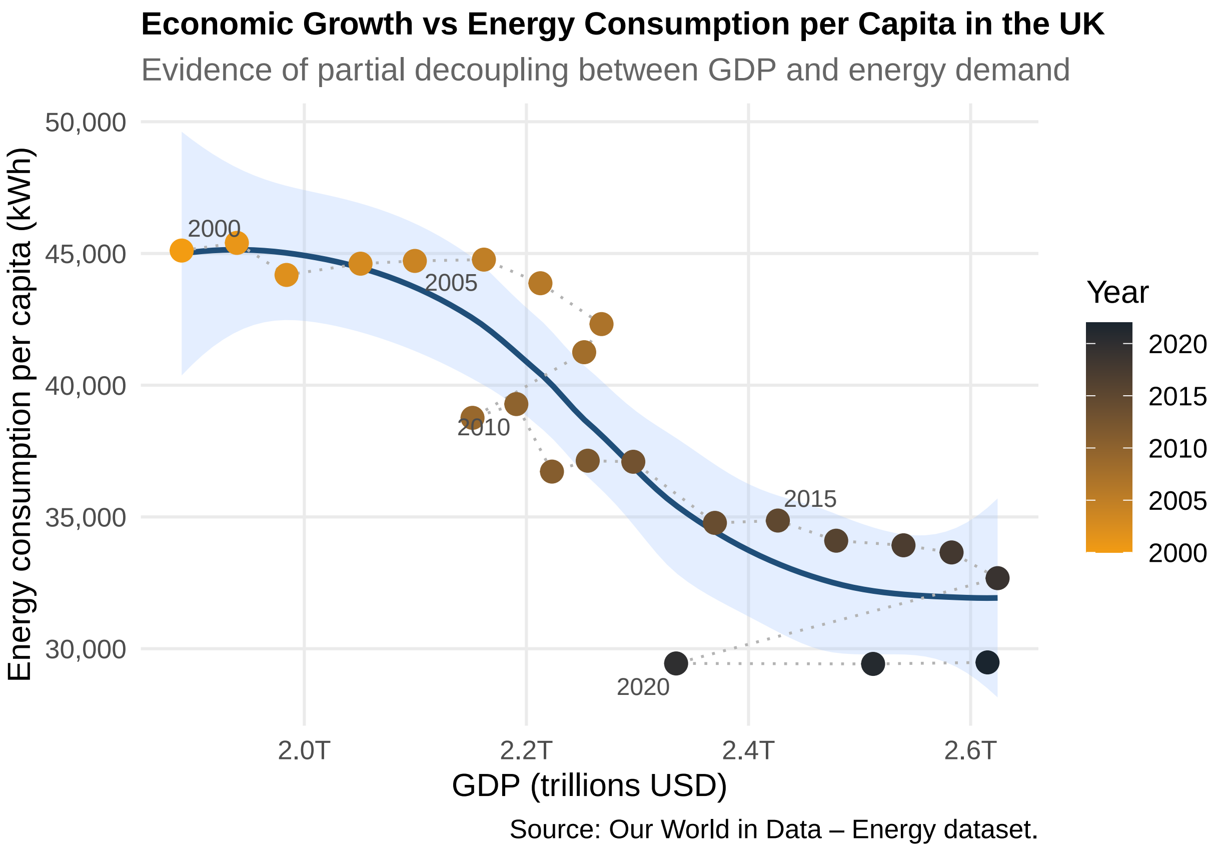
5.5 Economic Growth vs. Energy Consumption per Capita

This visualisation examines the relationship between economic growth and energy demand, allowing us to assess whether the UK has achieved a decoupling of GDP from energy consumption. It moves beyond electricity generation trends to address a broader structural question: whether economic growth can be sustained without a corresponding increase in energy use.

```r
label_years <- seq(2000, 2025, 5)

uk_decouple <- uk_electricity %>%
  filter(!is.na(gdp), !is.na(energy_per_capita)) %>%
  mutate(label = if_else(year %in% label_years,
                         as.character(year), NA_character_))

ggplot(uk_decouple, aes(x = gdp, y = energy_per_capita)) +
  geom_smooth(method = "loess", se = TRUE,
              colour = "#1f4e79",
              fill = "#a6c8ff",
              alpha = 0.3,
              linewidth = 1) +
  geom_path(colour = "grey70", linewidth = 0.5, linetype = "dotted") +
  geom_point(aes(colour = year), size = 3.5) +
  geom_text_repel(aes(label = label),
                  size = 3,
                  colour = "grey30",
                  box.padding = 0.3,
                  max.overlaps = 10) +
  scale_colour_gradient(low = "#f39c12", high = "#1a252f", name = "Year") +
  scale_x_continuous(labels = scales::label_number(scale = 1e-12, suffix = "T")) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(
    title    = "Economic Growth vs Energy Consumption per Capita in the UK",
    subtitle = "Evidence of partial decoupling between GDP and energy demand",
    x        = "GDP (trillions USD)",
    y        = "Energy consumption per capita (kWh)",
    caption  = "Source: Our World in Data - Energy dataset."
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title       = element_text(face = "bold", size = 12),
    plot.subtitle    = element_text(colour = "grey40"),
    panel.grid.minor = element_blank(),
    legend.position  = "right"
  )
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## (`geom_text_repel()`).
```



The relationship between GDP and energy consumption per capita suggests that the UK has made significant progress toward decoupling economic growth from energy demand. Traditionally, economic expansion was closely linked to rising energy use. However, the plot shows a different pattern: as GDP increases from around \$2.0 trillion to over \$2.6 trillion, energy consumption per person falls from roughly 45,000 kWh to below 30,000 kWh.

This downward trend indicates that the UK is producing more economic output while using less energy per person. This is often described as "decoupling," and it likely reflects a combination of factors, including improvements in energy efficiency and a shift away from energy-intensive industries toward a more service-based economy.

The LOESS curve provides additional insight into how this change has unfolded over time. Between roughly 2005 and 2015, the decline in energy use per capita becomes noticeably steeper, suggesting a period of rapid efficiency gains or structural change. After this period, the curve begins to flatten, indicating that reductions in energy consumption continue but at a slower pace. This may suggest that the UK is approaching a lower baseline level of energy use, where further reductions become more difficult without major technological or behavioural changes.

Overall, this pattern supports the idea that the UK's decarbonisation is not only driven by changes in the electricity mix but also by a reduction in overall energy demand. However, this result should be interpreted with some caution, as part of the decline may also reflect structural changes such as de-industrialisation rather than purely efficiency improvements.

6 Conclusion

```r
kable(
  data.frame(
    `Research Question` = c(
      "RQ1 - Fossil fuel electricity",
      "RQ2 - Low-carbon growth",
      "RQ3 - Decarbonisation outcomes"
    ),
    `Key Finding` = c(
      "Coal share collapsed from ~30% to under 1% by 2025, driven by the carbon price floor after 2013.
      "Low-carbon sources surpassed fossil fuels around 2017. Wind is the primary driver, rising from no
      "Carbon intensity fell from 522 to 217 gCO /kWh, a reduction of over 58%. The strong linear relati
    )
  ),
  caption = "Table 3. Summary of findings by research question.",
  col.names = c("Research Question", "Key Finding")
)
```

Table 3: Table 3. Summary of findings by research question.

Research Question	Key Finding
RQ1 – Fossil fuel electricity	Coal share collapsed from ~30% to under 1% by 2025, driven by the carbon price floor after 2013. Gas replaced coal as the dominant fossil fuel and persists at ~30% of generation, the most significant remaining structural barrier.
RQ2 – Low-carbon growth	Low-carbon sources surpassed fossil fuels around 2017. Wind is the primary driver, rising from near zero to over 25%. Nuclear has declined over the period, partially offsetting renewable gains. Solar has grown rapidly but from a low base.
RQ3 – Decarbonisation outcomes	Carbon intensity fell from 522 to 217 gCO /kWh, a reduction of over 58%. The strong linear relationship between fossil share and carbon intensity across all 26 years suggests this as structural decarbonisation rather than cyclical variation.

The UK's electricity transition over the past 26 years represents one of the most significant examples of power sector decarbonisation among developed economies. The near-elimination of coal, the rapid expansion of renewable energy, particularly wind, and the substantial reduction in carbon intensity all point to a genuine structural transformation rather than short-term or cyclical change.

However, this progress is not without limitations. First, natural gas remains embedded in the system as a key source of dispatchable power. While it has played an important role in enabling the transition away from coal, its continued presence highlights an ongoing dependency that exposes the UK to price volatility and delays full decarbonisation. Second, the gradual decline in nuclear generation represents an important counter-trend. As a stable low-carbon source, its reduction has limited the overall pace of emissions reduction and increased reliance on more variable renewable sources.

Finally, while the data suggests a degree of decoupling between economic growth and energy consumption, this should be interpreted cautiously. Part of the observed decline in energy use per capita may reflect structural changes in the economy, such as deindustrialisation, rather than purely improvements in efficiency. This indicates that progress on the demand side may be partially overstated when viewed in aggregate.

Overall, the UK has made substantial progress in transforming its electricity system, demonstrating that rapid decarbonisation is achievable with the right combination of policy, technology, and market conditions.

However, the remaining challenge lies in addressing residual dependence on natural gas and ensuring system reliability in a grid increasingly dominated by variable renewable energy. Achieving this will require continued investment in energy storage, grid flexibility, and other low-carbon technologies to support the next phase of the transition. In this context, the findings of this analysis contribute to broader global efforts aligned with Sustainable Development Goals such as SDG 7 (Affordable and Clean Energy) and SDG 13 (Climate Action).

7 Recommendation:

1. **Break down the low-carbon share further**: It would be useful to separate offshore and onshore wind to better understand how much of the growth is linked to government support and subsidy policies.
2. **Take a closer look at gas dependency**: Running a simple scenario analysis on different timelines for phasing out gas could give a clearer picture of how carbon intensity might change in the future.
3. **Compare the UK with other countries**: Looking at countries like France (which relies heavily on nuclear), Germany (with its Energiewende), and Denmark (a leader in wind energy) would help put the UK's progress into context.
4. **Consider grid stability, not just emissions**: Including things like curtailment, interconnector flows, and battery storage would give a more complete view of whether the system is not only cleaner, but also reliable.

8 References

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